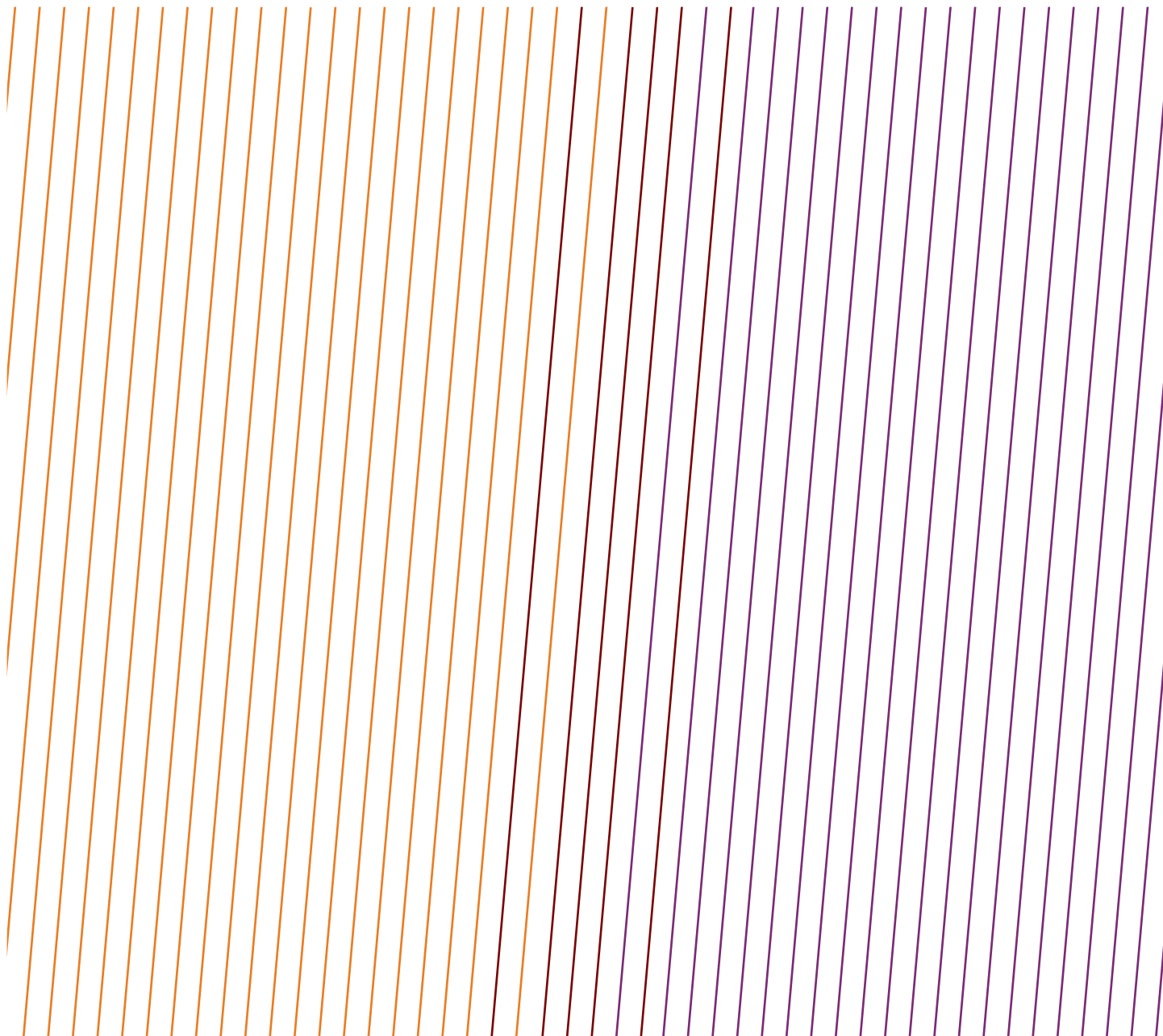


DATA SERIES

Safety performance indicators – Process safety events – 2025 data

Tier 1 PSE, fatal incident, and high potential event reports



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DATA SERIES

Safety performance indicators – Process safety events – 2025 data

Tier 1 PSE, fatal incident, and high potential event reports

Revision history		
VERSION	DATE	AMENDMENTS
1.0	July 2026	First release

Contents

TIER 1 PROCESS SAFETY EVENTS (PSE)	5
FATAL INCIDENTS CLASSIFIED AS TIER 1 PSE	124
FATAL TIER 1 PSE NOT INCLUDED IN THE SPECIFIC PSE DATA COLLECTION	130
FATAL INCIDENTS RELATED TO PROCESS SAFETY	131
HIGH POTENTIAL EVENTS CLASSIFIED AS TIER 1 PSE	132
HIGH POTENTIAL EVENTS RELATED TO PROCESS SAFETY	139

TIER 1 PROCESS SAFETY EVENTS (PSE)

Tier 1 PSE are predominantly lagging indicators related to Loss of Primary Containment (LOPC) referred to as a Process Safety Event (PSE). The Tier 1 Key Performance Indicator (KPI) records events with greater consequence within the four-tier approach.

For more information see IOGP Report 456 – *Process Safety – recommended practice on key performance indicators*. IOGP has been gathering Tier 1 PSE narrative reports from its Members since 2013. These include Tier 1 PSE reported both by companies and contractors. Event reports are categorized by region, country, location (onshore/offshore), cause, activity at the time of the event, and, from 2019 onwards, point of release.

The information provided here is not detailed; often the root cause of an incident cannot be established. However, the information should assist organizations to identify likely hazards and to question whether their own Safety Management System would have prevented the event occurring and mitigated its consequences.

The database from 2020 onwards is available and searchable at <https://data.iogp.org/ProcessSafety/Tier1PSE>.

Note that a descriptive report has not been provided for every Tier 1 PSE reported.

This database is a tool for learning and should not be considered a complete record of Tier 1 PSE in the upstream oil and gas industry or the IOGP Membership.

AFRICA ONSHORE

DATE: 14 Mar 2025

COUNTRY: Equatorial Guinea

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow

PROCESS SAFETY FUNDAMENTAL: We stop if the unexpected occurs.

INCIDENT DESCRIPTION:

Operator was alerted of LEL levels in the parameter of the condensate tank. It was identified that the tank was overflowing condensate.

WHAT WENT WRONG?

Level indication incorrect calibration.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Human Barrier Failures: Response to process alarm and upset conditions (e.g. outside safe envelope)

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

DATE: 09 Nov 2025

COUNTRY: Nigeria

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Sump/pit overflow

PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

On 9th November 2025 at about 1427hrs during a routine draining operation, there was loss of primary containment at the saver pit that went into the slot jetty area. Immediate action taken:

1. Draining stopped.
2. The outlet sluice gate was immediately closed.
3. A containment boom deployed at the slot jetty.

WHAT WENT WRONG?

The outlet sluice gate of the last line of defence pit inadvertently left open.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Ensure continuous monitoring during draining operations.

BARRIERS:

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

AFRICA OFFSHORE

DATE: 27 Jul 2025

COUNTRY: Angola

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Unspecified - other

MODE OF OPERATION: Normal

POINT OF RELEASE: Subsea: Subsea pipeline/flowline

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

At approximately 11:50 AM, a gas release was observed in the vicinity. The incident was reported by the surfer boat Captain who observed visible gas bubbles breaking the sea surface along the pipelines route.

Subsea team started a sonar survey to locate any abnormalities on the gas lines in the area. The Subsea team widened the survey area and focused on 8" gas line and detected a gas release from this line which was confirmed to be the source of the observed bubbling.

WHAT WENT WRONG?

Subsea gas leak was discovered after survey.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Area superintendent should form a cross-functional team to assess practicality and cost-effectiveness and prioritize repairs or process adjustments, including reinstating pigging capabilities and implementing alternative integrity assessment methods such as hydro testing and internal and external corrosion direct assessment - ICDA and ECDA, in line with company guidelines. Sept 30th, 2026.
2. Area superintendent to engage and set clear expectations to Operations frontline leaders and preservation coordinator to strictly follow and meet the requirements of the existing Fixed Equipment Inventory Guideline and preservation procedures. These documents are held by preservation team.
 - Involve operations and preservation coordinator to develop the preservation protocol for the identified equipment and ensure proper MOC is documented for the equipment placed out of service.
 - Consider involving Process Safety, Reliability, and Integrity - HCEPO (High Consequence Equipment Program Owner) positions in the communication of any change in equipment status.
 - Consider involving chemical SME on chemical requirements for preservation. Mar 31st, 2026.
3. Chemical specialist should verify that chemical treatment deficiencies are addressed through the new contract, ensuring products match fluid characteristics, and consider multiple contracts to overcome current supplier limitations. Due date: Mar 31st, 2026.
4. Asset integrity pipelines & corrosion team lead to develop Inspection, testing preventive maintenance (ITPMs) for piggable and unpiggable pipelines, and dashboards for tracking and monitoring the KPIs from the fluid sampling and corrosion monitoring information. ITPM should define:

- integrity critical tasks such as fluid sampling, chemical treatment, corrosion monitoring.
- The history and operating conditions of the pipeline
- The frequency of the execution of the ICTs for each pipeline.
- Update risk profile/ dashboard according to operational changes, as the inspections are performed. Sept 30th, 2026.

5. The Asset Integrity Pipeline & Corrosion team should engage Operations and Measurement/lab teams to identify lab equipment needs, asset integrity access to lab data base to facilitate direct data collection and set expectations to maintain use of the archive for technical documentation. Mar 31st, 2026.

BARRIERS:

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

DATE: 28 Feb 2025

COUNTRY: Angola

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Other production

POINT OF RELEASE: Wells, drilling and intervention: Well

PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

On February 28, 2025, around 9:00am field Operators, upon arrival at the field, spotted a gas leak bubbling a few meters from the well jackets. No oil sheen noticed, only gas bubbles. Later we found out to be related to a temporary abandoned well.

WHAT WENT WRONG?

Leak from abandoned well.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

NONE: Unspecified

DATE: 20 May 2025

COUNTRY: Angola

FUNCTION: Production

NUMBER OF DEATHS: 4

ACTIVITY: Workover/Well services

MODE OF OPERATION: Planned shutdown

POINT OF RELEASE: Breaking containment location

PROCESS SAFETY FUNDAMENTAL: We maintain safe isolation.

INCIDENT DESCRIPTION:

At approximately 03:10 AM, crews were performing flange management on the water 16" and 18" lines and installation of the blind flange on the 24" overboard line for hydrotest. During the process, crew encountered a clash with an abandoned open 18" line and started hot cutting the 18" line to make room to install a blind flange. Fire broke out at worksite, resulting in 17 people injured and 1 person missing.

WHAT WENT WRONG?

Hot cutting an abandoned line to install a blind flange led to a fire.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Shutdown Systems - including operational well isolation and drilling well control equipment

CAUSAL FACTORS:

NONE: Unspecified

DATE: 28 Nov 2025

COUNTRY: Angola

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Relief, Vent and Discharge Systems: Discharge to sea

PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

During normal operations, an oil level regulation valve suddenly closed.

This event led to instabilities in both the production and produced water networks.

These instabilities led to a deterioration in the quality of the treated produced water, part of which is continuously discharged into the sea.

An oil sheen was observed at sea, with an estimated volume of approximately 2m³ of oil.

WHAT WENT WRONG?

1. Positioner last inspection 2019 (degraded Positioner)
2. Unreliability of the LDT- 3510/3511 (DP)
3. Overdue calibration of Capacitive LDT1- 3510 (since 2023)

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Calibration and preventive maintenance should be implemented.

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 04 Sep 2025

COUNTRY: Angola

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)

PROCESS SAFETY FUNDAMENTAL: We recognize change.

INCIDENT DESCRIPTION:

On 04 September 2025, an uncontrolled release of approximately 1,064 kg of hydrocarbon and inert gas mixture occurred from the cargo tank vent lines on an FPSO, resulting in Emergency Shutdown activation. The release was caused by valves 64HCV0001 and 64HCV0002 opening unexpectedly during a reboot of controller.

WHAT WENT WRONG?

Controller failure was due to a firmware bug.

No formal agreement with supplier on timelines for firmware fixes or release notes during commissioning.

A software backup was performed during cold venting activities without prior communication between the ICSS FSR and the operations team. Additionally, the ICSS FSR did not verify the card restart configuration before generating and uploading the backup. The controller's configuration module was set to return to its "last saved" value upon restart when redundancy was not available.

Daily check for monitoring overall controller status was not established at this stage (Commissioning + Operation).

CROs not focus on system alarms at this stage (Commissioning + Operation).

Onboard team relied on verbal risk assessment. MOC "Deviation" exists, but triggers unclear for abnormal conditions.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Complete the firmware upgrade on the controllers.
- Develop and formalize a clear process that defines timelines and responsibilities for execution of anomalies and/or safety alerts.
- Conduct a comprehensive review of all controller module parameter configurations to confirm the safest restart settings are applied.
- Establish procedure to manage software changes during commissioning.
- For future projects, in alignment with the established process philosophy and specifically for systems that may result in unintended hydrocarbon releases to the atmosphere, it is essential to verify and define the software configurations during the engineering phase. These settings should ensure that actuated valves assume the safest possible position during restart conditions.
- Develop and implement a daily controller status verification checklist for the DCS Technician detailing specific items to check, timing, and completion steps.
- Improve and develop prioritization of the alarms for supporting control room operators' activities.

- Provide training and awareness sessions for relevant onboard personnel on how and when to conduct Deviations. This will build competence and ensure that risk assessments are properly recorded.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

DATE: 08 Nov 2025

COUNTRY: Nigeria

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Start-up

POINT OF RELEASE: Unknown/insufficient information

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

At approximately 0523 hrs. on November 8, 2025, an unusual sound was observed at the facility. Upon investigation, operations identified a fire on the upper section of the Auto-Thermal Reformer Unit 1 (ATR1). The Emergency Response Team (ERT) was immediately deployed to the location. In accordance with the plant's safety and control systems, ATR Units 1 and 2 as well as ASU Units 1 and 2 underwent automatic shutdowns due to inter-unit protective logic. The fire was fully extinguished by approximately 0630 hrs. There was no injury to personnel.

WHAT WENT WRONG?

Operating limits and equipment reliability.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

IR cameras previously installed had no capability for tracking and alerting operations of temperature changes.

The Isolation valve between reforming section and cooling train opened on loss of instrument air, contrary to its Fail-Last-Open (FLO) design intent. Latent functional defects went undetected due to the absence of a robust fail-safe validation program.

BARRIERS:

Hardware Barrier Failures: Ignition Control

CAUSAL FACTORS:

NONE: Unspecified

ASIA/AUSTRALASIA ONSHORE

DATE: 23 Sep 2025

COUNTRY: Bangladesh

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Unspecified - other

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline

PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

23 Sep 2025, 2:58 AM: Pressure drop (~6 psi) was detected during condensate shipping. Operations halted and security alerted.

Early Response:

BY Security Control Room (BYSCR) coordinated with Quick Response Team (QRT), local police, and pipeline officer. Foot patrols conducted (6:00–9:45 AM). No abnormality observed; shipping resumed at 11:50 AM after GTCL confirmation.

Community Alert & Spill Confirmation (5:15–5:37 PM):

Community reported condensate smell: shipping stopped. QRT confirmed spill at site.

Escalation:

5:38 PM: Incident Management Team (IMT) activated; Emergency Response Team (ERT 1) formed at 6:20 PM.

7:04 PM: Significant spill spread to low-lying land and canal waterways. ERT 1 left the necessary preparation, tools/logistics, for incident site at 7:30 pm (2-hour journey).

9:05 PM: Fire breakout reported; ERT 1 diverted to extinguish fire at bridge.

9:10 PM: Fire Service & Civil Defense (FSCD) engaged; fire extinguished by 10:20 PM with joint effort.

Leak Source & Containment:

11:55 PM: Leak source identified (illegal tapping with ball valve); area blocked with heavy soil.

24 Sep, 12:55 PM: The area was secured, and containment measures were implemented. Eventually, the leak was plugged, and 22. bbl condensate was recovered. Approximately 173 bbl condensate spill occurred.

Third party injuries:

Media reported three community members critically burned during the fire: hospitalized.

Tragically, three members of a nearby community, despite intensive care, passed away (two on 28 Sep, one on 1 Oct). The incident resulted in environmental contamination, property damage, and third-party loss of life.

WHAT WENT WRONG?

Illegal tap.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Inadequate engineering safeguard to prevent pipeline intrusion

- No current effective technological means in place for pipeline surveillance.

2. Ineffective security patrolling

- Security patrolling did not detect any abnormalities across pipeline ROW during/before illegal tap or after control room detection of potential concern via observed condensate pipeline pressure drop with increased flow rate.

3. Illegal tap leading to LOC escalating up to fire and impacting community personnel was not adequately anticipated

- No formal process existed to confirm or rule out abnormal activity or illegal tapping as the cause of pipeline pressure drops before restarting shipping. SSERP and emergency drills revealed gaps in preparedness and response after a pipeline LOC fire, compounded by poor communication among Company, local authorities, and the community, leading to delayed risk mitigation and continued exposure to condensate and ignition hazards.

BARRIERS:

Human Barrier Failures: Response to emergencies

CAUSAL FACTORS:

NONE: Unspecified

DATE: 07 Jan 2025

COUNTRY: India

FUNCTION: Drilling and Completion Operations

NUMBER OF DEATHS: 1

ACTIVITY: Production operations

MODE OF OPERATION: Routine maintenance

POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (not at intended discharge location)

PROCESS SAFETY FUNDAMENTAL: We walk the line.

INCIDENT DESCRIPTION:

Fatal fire following premature restart during PSV maintenance.

WHAT WENT WRONG?

A premature restart of the system without verifying reinstatement of the PSV resulted in uncontrolled hydrocarbon release, gas accumulation, ignition from a modified adjacent burner opening, and a fatal fire incident.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Critical safety devices must never be removed or bypassed without ensuring positive mechanical isolation, structured restart control, and thorough hazard assessment, particularly in hydrocarbon environments with adjacent operating units.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: 31 Aug 2025

COUNTRY: Indonesia

FUNCTION: Drilling and Completion Operations

NUMBER OF DEATHS: 1

ACTIVITY: Workover/Well services

MODE OF OPERATION: Well flow testing

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Flexible hose/piping

PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

Fatality during Well Surveillance.

WHAT WENT WRONG?

Disconnected pressurized hose connection impact to personnel.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

The importance of inspection for third party owned equipment.

BARRIERS:

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

DATE: 09 Dec 2025

COUNTRY: Indonesia

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Pipeline operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline

PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

INCIDENT DESCRIPTION:

Export pipeline's 8" leak at valve station.

WHAT WENT WRONG?

On Dec 9th:

- At 03:35 CCR operator received information from Security Management Centre (SMC), that security personnel found hydrocarbon leak from valve shelter. CPF Lead Operator, Field Operators and OSCT team went to the valve station. CCR supervisor on duty notified PIC (FSO & CPF) & Incident Commander.
- At 05:21 Shut-in export pipeline to FSO.
- At 06:45 ERT & OSCT team arrived at the valve station and started to clean up oil spill.
- At 09:01 Shut-in Train A to prevent Crude Oil Storage Tank (COST) from Tank Top and over-pressure.
- At 16:52 Conducted manual digging & installation removal, found leak source at 8" riser connected to 20" export pipeline.
- 9 Dec at 16:00 - 10 Dec at 14:00 Clamp installation.
- 10 Dec 14:00 Leak stopped.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Effective security oversight and immediate report and effective emergency response plan.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 16 May 2025

COUNTRY: Indonesia

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Planned shutdown

POINT OF RELEASE: Breaking Containment Locations: Piping/valve (inadvertently left) open to atmosphere

PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

Bottom column experienced leaks at bleeder valve and caused a vapor cloud.

WHAT WENT WRONG?

Bleeder valve was left open.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Ensure valve administration control is implemented prior main valve opening.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: 05 Aug 2025

COUNTRY: Indonesia

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping material/tubing

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Fire due to leak from 14" incoming piping in absorber of CO2 removal unit.

WHAT WENT WRONG?

Internal corrosion and improper design of control room.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

The importance of inherently safer design implementation and good engineering practice specific to hazardous area and safety instrumented function; the importance of inspection and maintenance for critical equipment.

BARRIERS:

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Ignition Control

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

ASIA/AUSTRALASIA OFFSHORE

DATE: 08 May 2025

COUNTRY: Australia

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Unspecified production

POINT OF RELEASE: Subsea: Subsea pipeline/flowline

PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

During subsea flushing activities on flowline, an estimated volume ranging between 1.58 - 16.15m³ of liquid hydrocarbon was discharged to the marine environment. This resulted in an intermittent surface sheen on the ocean with an area of ~2x2km.

WHAT WENT WRONG?

The liquid hydrocarbon was most likely “slops” from the floating production storage and offloading unit which had used the flowline to dispose of “slops” downhole.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Cleaning methods need to be designed with careful scrutiny to ensure effectiveness, especially when disposing fluids downhole.
- Accuracy and completeness of cessation of production records are essential to inform decommissioning risks and execution methodology.

BARRIERS:

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 18 Sep 2025

COUNTRY: China

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Subsea: Subsea equipment

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

A loss of hydraulic pressure was detected on a subsea system and troubleshooting indicated a subsea hydraulic flying lead had disconnected. The impacted well was shut in, methanol supply was isolated, and the system was depressurized while further inspection and investigation continued.

WHAT WENT WRONG?

The investigation is ongoing. Initial inspection observed the securing pin had sheared and was recovered for laboratory examination to determine the failure mechanism and sequence.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Lessons learned will be confirmed once laboratory results and investigation findings are complete.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

NONE: Unspecified

DATE: 01 May 2025

COUNTRY: Indonesia

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Pipeline operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Subsea: Subsea pipeline/flowline

PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

Gas release due to full bore rupture 4" gas lift near platform.

WHAT WENT WRONG?

There was anomaly external force during anchoring job and marine activity.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

The importance of risk assessment during SIMOPS between operation and marine.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Risk assessment and control

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 19 Aug 2025

COUNTRY: Malaysia

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Instrumentation and small bore tubing

PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

INCIDENT DESCRIPTION:

A leak of Corrosion Inhibitor (CI) occurred at one of the remote wellhead platforms. The source of the leak was identified at the bleed ring plug of the storage tank's level transmitter, which had become dislodged and came off from the fitting. The leak was isolated, and an estimated 4,600 litres of CI was released from the tank.

WHAT WENT WRONG?

Bleed ring plug became dislodged and came off from the fitting.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

CI usage checks at the end of each shift are a good practice for early detection of leaks and CI pump availability. It is important to ensure instrumentation components with the right materials are designed/installed to avoid galvanic corrosion.

Design improvements from new projects should also be considered for existing facilities to enhance safety and system integrity.

Leverage existing work management processes to highlight issues with PM task lists.

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

DATE: 22 Oct 2025

COUNTRY: Malaysia

FUNCTION: Production

NUMBER OF DEATHS: 1

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping material/tubing

PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

A steam release from an auxiliary steam line 14" occurred resulted in three (3) smoke detectors at the air intake activated that subsequently triggered the General Alarm. The Emergency Response Team was immediately mobilised to the scene. At the time of release members of the Engine Room immediately responded when radio calls were not answered.

Deceased Person (Maintenance Operator) was found unconscious near the Marine Gas Oil (MGO) tank and distilled water pump. The DP was brought to FPSO Hospital and was confirmed with ISOS Doctor as deceased.

Incident investigation is underway with the scheduled completion date in April 2026.

WHAT WENT WRONG?

This incident was due to the following factors:

- a. Condensation-induced water hammer
- b. The failure of the end plate weldment
- c. The end plate was ejected and hit the DP on the right side of his head

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Gaps in system operation, monitoring, and competency management. This serves as an important reminder of the need for strong technical control and workforce awareness when operating steam systems.

This incident highlights the importance of ensuring all crew, especially new and existing personnel, are properly trained and aware of steam system risks such as steam hammering, and are competent in following approved operating procedures. Critical equipment including control valves, steam traps, and piping must be maintained in proper working condition and operated in automatic mode where required to prevent condensate build-up and system damage. Adequate temperature and pressure monitoring must be in place and closely observed so that early intervention can be taken during abnormal conditions. Operating procedures must be regularly reviewed and updated to include steam hammering mitigation measures, and the Management of Change (MoC) process must be strengthened to ensure that any system modifications are properly assessed, documented, and reflected in as-built drawings.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PEOPLE (ACTS): Use of Protective Methods: Disabled or removed guards, warning systems or safety devices

EUROPE ONSHORE

DATE: 17 Mar 2025

COUNTRY: Hungary

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

The puncture is on the pipeline between #6 and #7, approx. 100 m from the #6 collector. The powerfully flowing gas completely carried away the soil from the backbone pipeline. Since we were not able to identify which backbone pipeline had a puncture at first, we closed all wells at the #6 collecting station at 09:30. VO, TO, OK are running next to each other. After identifying the faulty pipeline, we started depressurizing the #7 collecting station. The leaked gas could be 10000 Nm³. The restart of the #6 wells is currently in progress. #6 daily production: Br: 685.4 m³/d N: 22.5 m³/d, OK: 13,700 Nm³/d, sg: 90,100 Nm³/d. From the #7 collector, the OK gas is distributed towards the station, through which it arrives at the gas plant.

WHAT WENT WRONG?

A loss of primary containment occurred due to a puncture in the gathering pipeline, resulting in an uncontrolled release of flammable natural gas. The initial inability to immediately identify the exact damaged line delayed targeted isolation, requiring full station shutdown to control the situation.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Rapid identification and isolation capability of interconnected gathering lines should be reviewed and improved.

Consider enhancing pipeline integrity monitoring and inspection to detect potential weaknesses before failure.

Emergency response actions (well shut-in and depressurization) were effective and prevented escalation or ignition; however, response time optimization should be evaluated.

Review layout and marking of parallel pipelines (VO, TO, OK) to improve fault localization in similar scenarios.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 31 May 2025

COUNTRY: Italy

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Breaking Containment Locations: Loading/unloading coupling

PROCESS SAFETY FUNDAMENTAL: We maintain safe isolation.

INCIDENT DESCRIPTION:

An incident occurred during the transfer of an amine solution from a truck to a storage tank. The transfer was performed using a corrugated flexible hose connected to the truck with a threaded coupling and to the tank with a metal clamp. The end of the transfer was verified by observing hose vibrations generated during nitrogen displacement operations. After completing the transfer, one operator closed the nitrogen valve used for pressurization toward the truck and visually confirmed closure of the amine discharge valve on the truck. The second operator entered the containment area to close the tank valves. After closing the needle valve located near the tank, while attempting to close the ball valve, the hose suddenly detached from the nozzle, causing an amine splash that hit the operator's face through a gap between the visor and the chin. While attempting to wash at the eyewash station, the operator rubbed his face and eyes with contaminated hands, worsening the injury. He was immediately assisted by a colleague, taken to the infirmary, and then transported to the emergency room.

WHAT WENT WRONG?

Design: the absence of a non-return valve in the tank isolation circuit was identified, although it was included in the reference P&ID. The NRV was designed to be installed between the two tank isolation valves. Its presence would have prevented the incident by avoiding amine backflow into the final section of the pipeline, caused by the tank head, during the operator's movement from the truck to the tank to perform valve closure and tank isolation.

Management of Change: A MoC was underway to replace the current metal clamp fastening system used to connect the corrugated hose to the storage tank connections with safer fittings connections. The MoC process had started two years earlier. Engineering activities had been completed, and implementation had been finalized for some tanks but not yet for the tank involved in the incident.

Operational Procedure Misalignment: The existing loading procedure had been issued for a future plant configuration, as it was based on the implementation of the above-mentioned MoC, and therefore did not reflect the current operating conditions of the tank involved. Additionally, although the tank-side valve closing sequence was correctly performed by the operator, it did not include a clear and standardized sequence, generating confusion among operators due to the absence of a standard valve closure procedure.

Emergency Response Training: Insufficient awareness of decontamination procedures was identified. The eye injury occurred when the operator rubbed his eyes with contaminated hands.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Ensure technical/operational integrity: Verify that all design specifications are met during construction and installation. Address safety operational improvements deviations promptly through a completed MoC process and always ensure dedicated risk assessment which identifies possible mitigation measures to be implemented.

Keep procedures current: Operating procedures must reflect actual plant conditions, with clear roles, responsibilities, and standardized sequences

Strengthen emergency preparedness: Conduct regular training and drills for chemical exposure scenarios, including loading/unloading operations, to reinforce response protocols.

BARRIERS:

Human Barrier Failures: Response to emergencies

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: 09 Oct 2025

COUNTRY: Romania

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Other production

POINT OF RELEASE: Breaking Containment Locations: Loading/unloading coupling

PROCESS SAFETY FUNDAMENTAL: We control ignition sources.

INCIDENT DESCRIPTION:

After completing the loading of a vacuum truck from the pit located on the premises, a fire broke out between the shed and the truck cabin while the engine was running. The fire affected the truck cabin, and its driver and production operator suffered burns to the head.

WHAT WENT WRONG?

As the final investigation report is not available, no causes can be named yet.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

As the final investigation report is not available, no lessons learned issued yet.

BARRIERS:

Management System Element Barrier Failure: Risk assessment and control

CAUSAL FACTORS:

NONE: None: Unspecified

DATE: 01 Jan 1900

COUNTRY: UK

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Start-up

POINT OF RELEASE: Breaking Containment Locations: Piping/valve (inadvertently left) open to atmosphere

PROCESS SAFETY FUNDAMENTAL: We walk the line.

INCIDENT DESCRIPTION:

Loss of Primary Containment – Methanol Release from T-67114

WHAT WENT WRONG?

Two vent valves were found to be locked in the open position but should have been locked closed.

It was identified during log routines that there had been a significant drop in tank level T-67114 over an 11-hour period. This was subsequently investigated in the field and MeOH was found to be releasing into the tank bund from a filter vent line. Around 2 inches of liquid was found to be in the bund floor. On discovery of this, an attempt to

manually activate the foam deluge failed and released water into the bund, which agitated the methanol, causing several fixed gas detectors in the banded area to alarm. The site siren automatically sounded, and Emergency Response procedures were initiated.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Complete Line-Outs.

Control the Management of LO/LC Valves.

Change abnormal trends.

Verify emergency systems.

There has been an in depth thorough investigation into this event.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)

EUROPE OFFSHORE

DATE: 23 Sep 2025

COUNTRY: Norway

FUNCTION: Drilling and Completion Operations

NUMBER OF DEATHS: 0

ACTIVITY: Workover/Well services

MODE OF OPERATION: Abandonment

POINT OF RELEASE: Wells, drilling and intervention: Subsea well

PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

During permanent plugging, the 13 3/8" casing was cut with open BOP at 510 m after all barriers had towards all flowing units had been established below cutting depth. A significant volume of trapped gas was released and entered the rig floor and shaker area triggering gas alarms and ESD sequences. Diverter and annular were then closed and well secured.

WHAT WENT WRONG?

Casing cut operation assessed to have no risk of HC, based on presence of annulus cement and no flow units in overburden. Confirmed same by cement log interpretation, indicating fluid filled annulus. Cut casing with open BOP and light fluid (fresh water).

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Significant volume of trapped gas in annulus can be present even when barriers vs. all flow units have been verified. Gas migration can occur before annulus barriers have been fully established.

- Removing barrier elements shallow in the well cannot rely on timely activation of secondary barrier (BOP).
- Gaps identified and addressed in service supplier's internal procedure for log processing.

BARRIERS:

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 14 Feb 2025

COUNTRY: Norway

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Breaking Containment Locations: Loading/unloading coupling

PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

While performing maintenance to refill the lubrication oil reservoir, the operator disassembled the camlock coupling on the filling nozzle. During this process, oil unexpectedly sprayed back from the system.

The tank level dropped rapidly, prompting the operator to activate the emergency stop for the turbine to prevent further escalation. Two people were exposed to oil splashes and oil vapors.

WHAT WENT WRONG?

The demister in the vent line from the turbine lubrication oil tank was clogged with contamination, causing unexpected pressure and oil backflow during maintenance.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Filling nozzles to be equipped with shut-off valves for personnel protection

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 31 Mar 2025

COUNTRY: Norway

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Flexible hose/piping

PROCESS SAFETY FUNDAMENTAL: We walk the line.

INCIDENT DESCRIPTION:

Two operators were preparing the well (gas lift) for maintenance. During preparation, the operators were supposed to bleed down the pressure on an isolated segment. Communication was made with the control room, and the pressure on the equalize manifold was requested. It was then communicated over the radio that the equalize manifold was opened to the flare. The IP understood this as the entire segment being open for bleeding to the flare, while in reality, only the equalize manifold was open. This resulted in a limited pipe segment remaining without its pressure being bled off. The IP then opened a hose with a mounted "manometer manifold" at the end to prepare for N2 purging. The hose was lying on the deck. As the IP stepped back over the "manometer manifold," the hose ruptured, and the IP was struck by pressure and the hose on both legs.

WHAT WENT WRONG?

Undesired behaviour (internal and external) - Careless/unfocused/not observant/distracted. Bleeding to the hose was opened without the segment being bled to the flare. The hose was closed at the end, and the hose therefore experienced a pressure that exceeded the design pressure.

Work practices and accomplishment – Poor understanding of risk or misjudgement of potential hazard (Lack of pressure relief to flare).

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Conduct pre-job conversations to clarify roles and responsibilities.

Review and reinforce safe practices for hose usage.

Provide training on work processes.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

MIDDLE EAST ONSHORE

DATE: 28 Jan 2025

COUNTRY: Bahrain

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Pipeline operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping material/tubing

PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

Gas Lift Line had a major leak due to external corrosion – MGI Header also had a minor leak as a result of being impacted from the leak of the Gas Lift Line.

WHAT WENT WRONG?

1. Underground elbow failed due to external corrosion resulting from failure of dissimilar wrapping joint.
2. The 5-year PM failed to identify the failed wrapping and resulting external corrosion.
3. MGI Header-03 failed due to external impact from GL Header-02's leak.
4. Gas Lift Header-02 was not taken out of service in spite of having points under Alarm 3 thickness.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Drills are important for Operations/EMT Readiness
- Operations team must be trained on SCABA and provided with it to ensure timely emergency response around the clock.
- Safety measures and isolation shall take precedence over leaving a leaking line operational to avoid production loss.
- Handheld radios "Walkie-Talkies" to be put back in use for effective communication, especially during emergencies.
- Importance of officially raising critical conditions that may lead to incidents to the proper level in a timely manner.

- Improving procedures that have become accepted as a norm without enough scrutiny and highlighting shortcomings to management regardless of difficulty to implement the solution.
- Taking time to research from multiple credible sources before adopting new practices that could have detrimental impacts to process safety.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

NONE: Unspecified

DATE: 12 Nov 2025

COUNTRY: Iraq

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

- Leak first detected at 17:19 on 12th Nov.
- Line isolated at 18:25 on 12th Nov and start depressurizing through burn pit.
- Line at zero pressure @ 06:00 on 13th Nov

while leak no personal were in field.

WHAT WENT WRONG?

Gas leak from underground pipeline.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

-

BARRIERS:

Unspecified: Unspecified

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 26 Aug 2025

COUNTRY: Kurdistan Region Of Iraq

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow

PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

During a routine condensate tanker loading operation at Loading Bay, an overfill occurred resulting in an estimated loss of 1040 kg of condensate. The event resulted from the failure of the High-High level switch, which did not initiate the automatic system shutdown as designed. The abnormal condition was identified through CCTV monitoring, which revealed the product overflow. Immediate intervention was undertaken by activating the emergency shutdown push button, thereby halting loading operations.

WHAT WENT WRONG?

1. Failure of Primary Level Control Barrier- high-high level switch failed to operate and did not initiate the automatic closure of the associated loading valves (XV and SDV).
2. Design Flaws in Timer-Based Overfill Protection - The tertiary overfill protection relied on a preset loading timer based on a fixed batch concept, which did not account for varying tanker sizes or flow rates.
3. Redundant Barrier Not Maintained in Practice - A local high-level switch, previously identified as a safety barrier in earlier designs and referenced in the Control and Operating Philosophy, was retained physically but removed as a functional barrier during the re-HAZOP.
4. Procedural and Human Factors Weaknesses - Although pre-loading checks are required, they did not adequately account for the degraded condition of key safeguards.

The dependency on a single barrier and the lack of operator awareness about available fallback systems further compounded the risk.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Remove the logic for the end of sequence timer protection and remove its reference from the Control and Philosophy procedure.
2. Carry out an ergonomic design assessment of the Hatch Cover assembly.
3. Incorporate the local high-level Alarm as the secondary barrier.
4. Conduct a comprehensive Safety Critical Task Analysis's (SCTA) of the NGL road tanker loading operation.
5. Develop detailed Standard Operating Procedures (SOP) for the NGL/LPG and tanker loading process.
6. Deliver a training program to all affected stakeholders.

BARRIERS:

Hardware Barrier Failures: Detection Systems

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 20 Feb 2025

COUNTRY: Kuwait

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Crude oil leak from a 6" line coming from oil wells and going to a facility. The line was isolated. The total volume spilled is 20bbbls.

WHAT WENT WRONG?

Corrosion/erosion.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Periodic maintenance/inspection/testing.

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 24 Feb 2025

COUNTRY: Kuwait

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Crude oil leak from a 6" underground line coming from oil well and going to Gathering center. The line was isolated. The total volume spilled is 10 bbls.

WHAT WENT WRONG?

Corrosion/erosion.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Periodic maintenance/inspection/testing.

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 09 Nov 2025

COUNTRY: Kuwait

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Crude oil leak from a line coming from T-Junction and going central processing unit. The line was isolated. The total volume spilled is 10 bbls.

WHAT WENT WRONG?

Corrosion/erosion.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Periodic maintenance/inspection/testing.

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 12 Aug 2025

COUNTRY: Oman

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Several gas-detection alarms were triggered at a remote manifold station during evening operations, followed shortly by a report from field personnel of a major leak accompanied by a loud noise. All associated wells were remotely shut in, and the affected lines were manually isolated.

WHAT WENT WRONG?

The Cathodic Protection of the Monolithic Isolation Joint (MIJ) that is part of the 8" test flowline at a remote manifold station dislodged, resulting in a gas release to atmosphere for about 3–4 minutes. The segment of the 8" test flowline including the MIJ, about ~7mtr long, was propelled and landed at approximately 180m away from the manifold.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Main lessons learned:

1. Criticality of QA/QC site inspections during critical activities and final inspections against approved design.
2. Criticality of vendor selections and technical qualification/technical bid evaluation process where relevant experience is submitted and verified in similar application.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 17 Jun 2025

COUNTRY: Qatar

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Instrumentation and small bore tubing

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Flammable hydrocarbon released due to 30-LT0012A (1/2inch nipple) detached.

WHAT WENT WRONG?

Instrument connection (1/2 inch nipple) detached during normal operation.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Nipple material as per the instrument hook up drawing to be SS 316 while the failed nipple reported as C.S.
2. Material analysis report indicated the failure root case is overloading associated with internal corrosion. Galvanic corrosion due to incorrect materials used in sour service.
3. PMI (Positive Material Identification) procedure does not Cover SS (Stainless Steel) components during major projects in 2003 for the Asset.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

DATE: 17 Apr 2025

COUNTRY: Qatar

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping material/tubing

PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

INCIDENT DESCRIPTION:

On 17 April 2025, at 09:15 AM, during plant round, an operator observed an abnormal sound. Upon further inspection of the area, a vapor cloud and condensate leak were identified in Unit-84 from the Slug Catcher Filter area, (2" Line at upstream of 84-BDV-01.19).

Shift Supervisor/MCR Panel were immediately informed about the leak. The operator pulled the MAC, activated the deluge system, and provided a water curtain as per the Shift Supervisor's direction. Plant emergency shutdown was initiated; the leaking section was isolated by closing ESVs and depressurized through the PCV on the Pre-Flash Drum to flare.

On inspection, a major hole, around 3" equivalent leak size was identified upstream of BDV01.19.

This leak triggered a Tier-2 Emergency Response, with personnel evacuating. 70 personnel were evaluated by the medical team, with toxic clips detecting 5ppm H2S at an occupied area, ~1.3Km away from the leak location. Of the personnel evaluated for gas exposure, 29 were classified as First Aid cases.

WHAT WENT WRONG?

1. Water breakout from offshore facilities leading to accelerated corrosion due to ingress of free water in the inlet facilities beyond its design limits.
2. Design parameters changes (water content) did not trigger changes to the Inspection Program.
3. Mismatches were identified between the Inspection report and QA/QC inspection report (QIR) for the repair in year 2021, thus the lower thickness section was not replaced. Inspection report didn't mention the dimension.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. On identification of operating/process parameter deviations, risks to the asset and stakeholders shall be identified and a risk management plan shall be instated.
2. A mechanism to effectively share the data between interacting assets shall be implemented.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

MIDDLE EAST OFFSHORE

DATE: 11 Oct 2025

COUNTRY: Israel

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Upset

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Valve (body, stem, plugs)

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

At 08:55 am on the 11th of October a line of sight gas detector located on the production mezz deck indicated High High gas at which point the Control Room Operator (CRO) activated the Fire & Gas alarm and all personnel were instructed to go to muster. Additional gas detectors indicating High on the production deck location at which point the OIM instructed the CRO to activate the Emergency Shut Down due to the number of detectors that had been activated on Production Mezz Deck and Production Deck.

WHAT WENT WRONG?

Gas detector displayed high pressure notating a leak.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Seal material is not suitable for the average and minimum operating temperatures encountered in service.

The trends suggest that the shutdown valve on the downstream side of the filter vessel starts closing slightly earlier than the shutdown valve upstream. This might be causing the pressure spike in the vessel.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

NONE: Unspecified

DATE: 17 Oct 2025

COUNTRY: Qatar

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Routine maintenance

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Valve (body, stem, plugs)

PROCESS SAFETY FUNDAMENTAL: We maintain safe isolation.

INCIDENT DESCRIPTION:

During a scheduled GLSDV function test, the Well head maintenance team confirmed normal operating conditions before detecting a hissing sound at the riser line while leaving the platform. The leak was immediately reported, and a manual Emergency Shutdown (ESD) was initiated from the boat landing, followed by safe crew evacuation. A subsequent inspection with SCBA confirmed all wells and riser SDVs had tripped, and additional isolations were applied.

WHAT WENT WRONG?

The cause was identified as a detached ½" grease nipple upstream of valve, which had been in long-term isolation due to prior oil seepage, with replacement already planned under the riser repair program.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

The investigation is yet to be completed.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 11 Mar 2025

COUNTRY: Qatar

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping material/tubing

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

On 11th March 2025 at 2030, a gas leak was reported from the wellhead platform Chemical Inject (CI) Quill. The leak continued for about 6 days due to the passing emergency shutdown valve. The hydrocarbon gas at 85 – 90 barg with 0.8 mol % H₂S leaked from a 3 mm weep hole.

WHAT WENT WRONG?

Weld crack on corrosion inhibitor quill, Inadequate design of the chemical Injection quill with regards to quill length and the wake frequency calculations.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

A specification shall be developed for selection of the quill and requirements for testing of the quill.

The lessons learnt database shall update on a regular basis to ensure lessons are thoroughly captured in lessons learnt data base.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

DATE: 31 May 2025

COUNTRY: Qatar

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Instrumentation and small bore tubing

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

On 31st May 2025 at 20:45hr, a loud jetting sound from heard from unmanned offshore platform. Emergency shutdown (ESD) was initiated from at 21:05 hr. The operating pressure of the isolated pressure differential transmitter (PDT) line decreased from 104 barg to 82 barg within 1 hour following ESD activation. The line with PDT was operating at 104 barg and 60°C with 0.8 mol% of H₂S. The static inventory of 7550kg depleted and pressure dropped from 104 barg to 0 barg over 15 hours.

WHAT WENT WRONG?

Two cover screws securing the PDT assembly had broken and detached, leading to LOPC through the gaps between the instrument tubing and the PDT diaphragm.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Do not ignore design requirements of small components during projects. Failures always start with the weakest elements and parts not focused on. Attention to details is important, especially for high pressure sour gas related equipment.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

DATE: 22 Sep 2025

COUNTRY: UAE

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Valve (body, stem, plugs)

PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

During this operation, the handle and stem assembly of the 3-way valve, operating under an oil pressure of 55 bar, detached from the valve body and struck one operator on the lower left side of the face, causing a facial injury. The assisting operator sustained mild facial swelling due to oil exposure to the face and left eye.

WHAT WENT WRONG?

The top event involving the detachment of the valve handle, stem and lifting jack nut, and subsequent injury to the operator is attributed to a weak inherent design of the valve, combined with the improper operation of the 3-way valve (procedural).

CORRECTIVE ACTIONS & RECOMMENDATIONS:

When operating high-pressure, safety-critical valves (like 3-way diverter valves), design vulnerabilities and stored-energy hazards must be controlled by strict procedural compliance and a robust asset management regime. Failing any one of these layers can turn a routine changeover into a serious line-of-fire event.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

No Causal Factors Allocated

NORTH AMERICA ONSHORE

DATE: 28 Jan 2025

COUNTRY: Canada

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Wells, drilling and intervention: Well

PROCESS SAFETY FUNDAMENTAL: We recognize change.

INCIDENT DESCRIPTION:

An H2S alarm was received at a facility, and operations responded to investigate. A wellhead was found venting sour gas, and the release was stopped by manually closing valves.

WHAT WENT WRONG?

A wellhead coupling that was not rated for the service failed. The work scope was changed and implemented without an effective deviation/stop-work process, and controls for sourcing/verification of wellhead components were not sufficient for the task.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Ensure well programs and return-to-service steps clearly specify required parts, ratings, and hold points, including formal notification/hand-off requirements between groups. Implement a controlled process for wellhead component sourcing/verification (e.g., managed through a controlled inventory pathway) and reinforce deviation/stop-work expectations when scope changes occur.

BARRIERS:

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 28 Feb 2025

COUNTRY: Canada

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Breaking Containment Locations: Loading/unloading coupling

PROCESS SAFETY FUNDAMENTAL: We maintain safe isolation.

INCIDENT DESCRIPTION:

During a fluid-hauling task, water transfer to a tank truck was completed and the hose was disconnected. A valve was found not fully closed, and hot fluid sprayed during subsequent valve manipulation. The job was stopped, emergency medical response was initiated, and the well/site was shut down to support response and control the release.

WHAT WENT WRONG?

Residual energy remained in the system at the time of disconnection/valve manipulation, resulting in a hot-fluid spray. The investigation identified gaps in procedural requirements for verifying zero energy and specifying chemical PPE for hot-fluid tasks, and noted that the system configuration did not readily allow field verification of “zero energy.”

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Update procedures and minimum PPE requirements for hot-fluid tasks (including defined thresholds) and incorporate explicit “verify zero energy” expectations. Where verification is not practicable, implement engineering/administrative improvements (e.g., bleed/verification points or safer valve/coupler configurations) so that residual energy can be confirmed and controlled before disconnecting equipment.

BARRIERS:

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PEOPLE (ACTS): Use of Protective Methods: Personal Protective Equipment not used or used improperly

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 07 May 2025

COUNTRY: Canada

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Wells, drilling and intervention: Well

PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

An uncontrolled release to surface occurred at a thermal well pad, and incident management resources were activated to support response and stabilization. The investigation is ongoing.

WHAT WENT WRONG?

The investigation is ongoing; indicated the event involved a well casing failure and the causal mechanism is still being confirmed.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Lessons learned will be finalized once the investigation is complete.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

NONE: Unspecified

DATE: 19 Aug 2025

COUNTRY: Canada

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline

PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

INCIDENT DESCRIPTION:

An operator identified abnormal pressure conditions on a CO2 injection system and confirmed a pipeline failure on a CO2 flexpipe segment. The release was isolated and response actions were initiated.

WHAT WENT WRONG?

The failure mechanism was associated with pipeline stressors under operating conditions, including thermal expansion, rapid pressurization/depressurization, and loss of restraint in wet soils.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Strengthen commissioning/operating practices to avoid rapid pressure cycling, mitigate seasonal flooding and loss of restraint, and apply enhanced leak detection to reduce release duration. Use repairs/hydrotesting and targeted integrity verification for adjacent segments exposed to similar conditions, and communicate learnings across similar assets.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 10 Mar 2025

COUNTRY: Canada

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Breaking Containment Locations: Sample system

PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

Operator noticed that the tap was in the open position. Release had overflowed onto the building floor. Small amounts of oil had leaked onto the ground outside the building.

WHAT WENT WRONG?

Valve incorrect position.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

DATE: 25 Mar 2025

COUNTRY: Canada

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Equipment: Pump

PROCESS SAFETY FUNDAMENTAL: We recognize change.

INCIDENT DESCRIPTION:

Operator observed oil spilling from tank.

WHAT WENT WRONG?

Seal failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 20 Sep 2025

COUNTRY: Canada

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Valve (body, stem, plugs)

PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

INCIDENT DESCRIPTION:

Release occurred from water dump bypass valve.

WHAT WENT WRONG?

Valve erosion.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 10 Dec 2025

COUNTRY: Mexico

FUNCTION: Production

NUMBER OF DEATHS: 1

ACTIVITY: Pipeline operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Valve (body, stem, plugs)

PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

The worker was holding the position of Trainee with a service company, while performing activities on the 18" floating gate valve located on the inlet line to the separator at the Gas Transportation Processing Centre, was struck in the front right hip by a metallic component. He was transported to the medical office located outside the facility, where he was attended by the on-duty physician, who declared him deceased at 10:30 hours.

WHAT WENT WRONG?

A Trainee was struck in the front right hip by a metallic component while performing work on an 18" floating gate valve, on the inlet line to the separator.

The impact indicates a loss of control of stored mechanical energy or unexpected movement/ejection of a component during valve-related activities.

Critical hazards associated with heavy components and pressurized systems were not effectively controlled at the moment of the task.

The worker sustained fatal injuries and was declared deceased by the attending physician at 10:30 hours.

The event suggests deficiencies in hazard identification, energy isolation (LOTO), mechanical restraints, exclusion zones, or task supervision, particularly considering the worker was a trainee.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Valve interventions on large-diameter systems are high-risk activities and require strict control of stored energy (pressure, mechanical, gravitational).

Full isolation, depressurization, verification of zero energy, and mechanical securing of components must be completed before any work begins.

Exclusion zones and line-of-fire controls are mandatory when working with heavy or potentially mobile metallic components.

Trainees must not be exposed to critical tasks without direct supervision, clear role definition, and confirmation of competency.

Permit to Work (PTW), Job Safety Analysis (JSA), and LOTO procedures must explicitly address:

- Component movement or ejection.
- Required restraints and tooling.
- Safe positioning of personnel.

Contractor safety management must ensure alignment with site standards, including training, supervision, and enforcement of stop-work authority.

Fatal events reinforce that engineering and procedural controls must never rely solely on experience or assumptions; verification is essential.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

DATE: 24 Feb 2025

COUNTRY: Mexico

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Routine maintenance

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

During a drone inspection along the Right of Way of the 16" Ø crude oil pipeline, a loss of containment was detected at km 31+812, which was subsequently confirmed by pipeline personnel, in a low-lying flood-prone area. Provisional repair work was initiated with the installation of a clamp; however, at 20:00 h it was determined that the clamp could not be installed because the assessment indicated a longitudinal weld failure of approximately 60 cm, caused by electric resistance welding, located at the 7 o'clock position.

WHAT WENT WRONG?

A loss of containment was detected during a drone inspection along the DDV of a 16" crude oil pipeline at km 31+812.

The affected section is located in a low-lying, flood-prone area, which likely accelerated corrosion and integrity degradation.

Initial mitigation attempted through a provisional clamp repair was unsuccessful.

Further assessment identified a longitudinal weld failure (~60 cm) at the 7 o'clock position, associated with electric resistance welding (ERW).

The nature and extent of the defect made clamp installation ineffective and unsafe, indicating a structural failure rather than a localized leak.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

ERW longitudinal welds represent a known integrity risk and require focused monitoring, especially in aging pipelines.

Flood-prone and low-lying areas significantly increase the likelihood of corrosion and must be treated as high-risk segments.

Drone inspections are effective for early detection, but must be followed by timely and thorough field verification.

Temporary repair methods (e.g., clamps) are not suitable for extensive weld failures; clear criteria must exist to determine when replacement or shutdown is required.

Integrity management programs should prioritize:

- Targeted inspection of ERW pipelines.
- Geohazard and environmental risk mapping.
- Preventive replacement or rehabilitation strategies.

Early identification prevented escalation, but the event highlights the need for proactive integrity assessments rather than reactive repairs.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 15 May 2025

COUNTRY: Mexico

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping material/tubing

PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

A leak was detected on the 3" Ø liquids line toward the battery limit, which is currently out of operation.

Actions: It was verified that the valves of the liquids circuit were closed; the area around the central rack where the line runs was barricaded, and maintenance and inspection personnel were notified for evaluation.

WHAT WENT WRONG?

A leak was detected on a 3" diameter liquids line near the battery limit, even though the line was reported as out of operation.

The presence of a leak on an idle line suggests residual liquids trapped in the system or incomplete isolation and draining.

Possible contributing factors include internal corrosion, degraded piping integrity, or valve leakage allowing residual fluids to remain.

While no escalation occurred, the leak confirms that out-of-service lines still pose a risk if not fully decommissioned or preserved.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Out-of-operation does not mean risk-free: lines must be fully isolated, drained, and verified empty to prevent leaks.

Residual fluids in idle systems should be considered during integrity and hazard assessments.

Barricading and access control were appropriate actions to protect personnel and prevent exposure.

Timely notification of maintenance and inspection teams supports effective evaluation and corrective action.

Integrity management programs should include:

- Inspection and monitoring of idle or rarely used lines.
- Clear criteria for mothballing, preservation, or permanent decommissioning.

Proper management of inactive equipment helps prevent unexpected releases and improves overall site safety.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 20 Sep 2025

COUNTRY: Mexico

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Pipeline operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

A loss of containment was detected on the Right of way of the 16" Ø oil-gas pipeline that runs from the #1 Separation Battery to the interconnection of the 16" Ø oil-gas pipeline (L2) at the AT of the #2 Separation Battery (km 2+169). Together with the operations personnel, operational adjustments were carried out to shut in the wells, proceeding to isolate the section of the 16" Ø oil-gas pipeline from #1 Separation Battery to the AT of #2 Separation Battery, taking the 16" Ø (L2) oil-gas pipeline out of service in order to begin a controlled depressurization.

WHAT WENT WRONG?

A loss of containment was detected on the DDV of the 16" Ø oil-gas pipeline at km 2+169, between the #1 Separation Battery and the interconnection with the 16" Ø (L2) oil-gas pipeline at the #2 Separation Battery.

The event indicates a failure of pipeline integrity, potentially due to corrosion, material degradation, mechanical damage, or weld/joint defects.

The release required operational intervention, demonstrating that the pipeline segment represented an active risk to safe operations.

The incident caused production impact, as multiple wells feeding the system had to be shut in to allow safe isolation and depressurization.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Prompt detection and coordinated operational response with operations personnel helped prevent escalation.

Immediate well shut-in, isolation, and controlled depressurization are essential actions following any oil-gas pipeline loss of containment.

DDVs and interconnection points are critical integrity locations and must be prioritized in inspection and monitoring programs.

Integrity management should emphasize:

- Regular inspection of oil-gas pipelines, especially near batteries and tie-ins
- Corrosion control and leak detection systems
- Production losses highlight the importance of preventive maintenance and early defect identification to avoid unplanned shutdowns.
- Clear procedures and communication between operations teams are key to safe and effective emergency response.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 30 Nov 2025

COUNTRY: Mexico

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Sump/pit overflow

PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

INCIDENT DESCRIPTION:

At 18:00 hrs on 11/30/2025, on-call personnel notified the Production Supervisor via radio about an event in the measurement tank (TM) area, detecting on site the passage of liquids in the suction valve from TM-1 to the RA-11 oily register, which caused an increase in levels in the API dam and RA-05 oily registers (separator area), RA-12 (measuring tank dam) and S/N registers in the area of pumps with hydrocarbon spillage on the ground of these areas, immediately an attempt was made to pump with the Internal Combustion Motor Pump 03 (MBCI-03) in a discontinuous manner due to the low suction pressure, so the suction valve of the TM-1 was dismantled. managing to contain the passage of liquid to the API dam and oily registers with diversion of oil to the TM containment dam and later pumping to the vertical tanks through the pump (MBCI-03).

WHAT WENT WRONG?

On the day of the event, Tank TM-1 had been dewatered by opening the suction valve towards its base, directing water to the API separator via the oily drains. During this operation the water was pumped from the API separator to the larger vertical storage tank TV-5. Once the water had been decanted the valve was closed.

Later in the afternoon an operator detected flow in the oily water system and determined that the valve on TM-1 was passing. Suspecting debris was lodged beneath the gate, the operator opened the valve to flush the debris and then closed it. This and other attempts to fully close the valve were unsuccessful and the operator decided to use a pipe wrench to close it. The force broke the lifting nut and there was a significant increase in flow.

Despite the pump running at the API separator and additional pumping of oil to the export system, the site was unable to prevent the API separator from being overwhelmed, and oil backflowed up the oil drainage system, overtopping covers in several locations to spill onto the concrete hardstanding. A flange was loosened at the suction valve on tank TM-1 and the remaining contents of the tank allowed to drain into its containment bund.

About 198bbls of oil was lost from the tank. Most was contained within the drains and the API separator and pumped to tank TV-5 or collected by a vacuum truck.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Update valve maintenance and inspection protocols. – Review and update valve maintenance protocols.

Factors to consider may include:

- means to identify rags debris and foreign matter during annual maintenance and repairs
- scheduled stripping and overhauling (or replacement) of critical valves, especially ageing ones
- incorporate borescope or visual inspection steps for valves during maintenance to detect internal contamination, material loss and internal condition
- oversight of maintenance contractor

- Management should actively collaborate with workers to identify opportunities to encourage more frequent pausing of work, seeking assistance, and escalating issues, and subsequently develop a plan to implement any recommended actions.
- Resurvey oil and rainwater drainage systems to confirm arrangements. Once completed review pollution prevention / environmental protection philosophy related to these, the secondary containment systems and API Separator and develop procedures to manage the systems and protect the environment. Roll out training, and supporting processes (such as testing and valve position logs for draining accumulated water) as well as emergency and recovery procedures (preventing offsite migration, safety hazards, post spill clean up etc)
- Halt any drainage of TM-1 and TM-2 to the oily drains until the risk has been reassessed and adequate isolation provided. Update drawings and MOC related to tank drainage and develop procedure or instruction for the activity.

BARRIERS:

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

DATE: 30 Jan 2025

COUNTRY: USA

FUNCTION: Drilling and Completion Operations

NUMBER OF DEATHS: 0

ACTIVITY: Workover/Well services

MODE OF OPERATION: Well intervention / Well servicing

POINT OF RELEASE: Wells, drilling and intervention: Well intervention equipment

PROCESS SAFETY FUNDAMENTAL: We walk the line.

INCIDENT DESCRIPTION:

While completing a frac, fluid released from a valve and the cellar area.

WHAT WENT WRONG?

Incorrect valve setup.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

DATE: 01 Feb 2025

COUNTRY: USA

FUNCTION: Drilling and Completion Operations

NUMBER OF DEATHS: 0

ACTIVITY: Drilling

MODE OF OPERATION: Drilling

POINT OF RELEASE: Wells, drilling and intervention: Well intervention equipment

PROCESS SAFETY FUNDAMENTAL: We stop if the unexpected occurs.

INCIDENT DESCRIPTION:

While pulling out of hole, swedge below the lubricator unexpectedly released.

WHAT WENT WRONG?

Connection failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

NONE: Unspecified

DATE: 16 Feb 2025

COUNTRY: USA

FUNCTION: Drilling and Completion Operations

NUMBER OF DEATHS: 0

ACTIVITY: Workover/Well services

MODE OF OPERATION: Well intervention / Well servicing

POINT OF RELEASE: Wells, drilling and intervention: Well intervention equipment

PROCESS SAFETY FUNDAMENTAL: We stop if the unexpected occurs.

INCIDENT DESCRIPTION:

While fracking, a fire occurred from a frac pump.

WHAT WENT WRONG?

Pump failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

NONE: Unspecified

DATE: 06 Apr 2025

COUNTRY: USA

FUNCTION: Drilling and Completion Operations

NUMBER OF DEATHS: 0

ACTIVITY: Workover/Well services

MODE OF OPERATION: Other production

POINT OF RELEASE: Wells, drilling and intervention: Well intervention equipment

PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

During tree-up operations on the well, surface equipment failed, resulting in a loss of containment and an injury to a contractor's employee. The injured person was transported to a medical facility for treatment.

WHAT WENT WRONG?

Surface equipment failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Landing joint was not made up properly prior to running landing tool. The Vault wellhead technician could not apply the adequate amount of torque with the make-up method used to properly make-up the running tool and landing joint.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

NONE: Unspecified

DATE: 09 May 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Unspecified - other

MODE OF OPERATION: Normal

POINT OF RELEASE: Equipment: Filter

PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

INCIDENT DESCRIPTION:

Based on witness statements and post incident evidence. On 5/9/2025, at approximately 12:30 hours, a gas leak occurred on the gas filter separator located at the Compressor Station. A vapor cloud formed until it contacted an unknown ignition source. The vapor cloud "exploded" and caused a fire which damaged the adjacent glycol reboiler along with other equipment.

WHAT WENT WRONG?

Believed to be a gasket failure on the gas filter separator.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Preventive maintenance, inspection, testing, or repair: Inadequate.

No inspection expectations: There were no defined expectations to inspect the quick opening closure (QOC) door alignment during PSSR, 10-year AI inspections, or preventive maintenance.

Post-incident findings: The manufacturer found the filter separator door was sagging and out of tolerance.

Signs of forced closure: Knocking marks on the latch assembly and lock-ring indicated the door had been forcibly closed.

Root Cause #2: Quality control or acceptance testing: Inadequate.

Seal integrity compromised: Seals and packaging did not meet manufacturer standards.

Improper storage: Seals are required to be stored in UV-protective bags, but suppliers removed and repackaged them individually.

Damaged new seal: A new seal still in its package was found coiled, kinked, and damaged.

BARRIERS:

Hardware Barrier Failures: Ignition Control

CAUSAL FACTORS:

NONE: Unspecified

DATE: 25 May 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Valve (body, stem, plugs)

PROCESS SAFETY FUNDAMENTAL: We maintain safe isolation.

INCIDENT DESCRIPTION:

Compressor blowdown led to a release of 251,284 Kg (9,289 MSCF) of natural gas into the atmosphere.

The event was classified a tier 1 LOC.

This event was caught by our methane flyover program.

WHAT WENT WRONG?

Unspecified.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Valve malfunction.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

NONE: Unspecified

DATE: 03 Jul 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

At approximately 3:15 pm on Thursday, July 3rd it was reported that a vacuum truck operator was skimming oil into the oil tank. During this process the oil tank started leaking out of the pinhole at the bottom of the silver 2nd oil tank.

WHAT WENT WRONG?

Pin hole leak noticed in oil tank.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Unknown.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

NONE: Unspecified

DATE: 22 Mar 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Unspecified - other

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow

PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

The water line was clogged with solids causing oil tanks to overflow. This resulted in a spill to a lined berm and adjacent pad of 68.5 bbls of crude oil, of which 66 bbls were recovered.

WHAT WENT WRONG?

Clogged line caused overflow of tank.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Unknown.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

NONE: Unspecified

DATE: 12 Sep 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Responding to volume anomalies, a Reliability Operator (RO) and Senior Reliability Operator (SRO) discovered surfaced emulsion in an offsite right-of-way. A 6" emulsion line leak was discovered on an emulsion line to a production facility offsite. Approximate dimensions of surface impacts is a l-shape stream with low lying pools 500 feet by 700 feet with varying widths in low spots (10'-100') traveling northwest from the line.

WHAT WENT WRONG?

Pipeline failure/malfunction.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

NONE: Unspecified

DATE: 18 Sep 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 1

ACTIVITY: Pipeline operations

MODE OF OPERATION: Routine maintenance

POINT OF RELEASE: Equipment: Pig launcher/receiver

PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

A contract crew was in the process of cleaning a sales oil line for new pad tie-in work as part of a pipeline infrastructure project when a crewmember was injured during work with a sales oil pig receiver resulting in a fatality.

WHAT WENT WRONG?

Worker was struck in the head.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

NONE: Unspecified

DATE: 12 Oct 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Unspecified - other

MODE OF OPERATION: Unspecified production

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Valve (body, stem, plugs)

PROCESS SAFETY FUNDAMENTAL: We stop if the unexpected occurs.

INCIDENT DESCRIPTION:

Crew were in the process of installing an injection quill/H2S analyser probe into a valve positioned on top of the sales gas line in the dehydration building. As part of the installation process they loosened a 1" bull plug positioned on top of the valve (facing straight up approx. 3-3.5 ft. from the ceiling), once the plug was loose gas started to release, worker exited the ladder opening the door to ventilate, decision was made to continue to loosen the plug to relieve what they assumed was excess pressure. As they continued to loosen the plug it released from the valve lodging itself in the ceiling of the building with gas releasing into the building the workers exited the building activating the ESD.

WHAT WENT WRONG?

Workers did not communicate schedule or plan with the company. Permit to work was not signed, approved, or followed. Walkdown was not completed. Work was not on the schedule.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate communication

DATE: 13 Jan 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Routine maintenance

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping joint

PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

Contractor repairing methanol pump caused 3/8" line to part, releasing gas; check valve and fitting failed; no injuries.

WHAT WENT WRONG?

On 1/13/25, a contractor was repairing a methanol pump in the compressor building when the 3/8" line parted at the line connection. The contractor isolated the line between the manifold and the ball valve to check flow of chemical to the pump. The contractor reinstalled the line and the line failed causing the gas release. It was found that the check valve on the compressors 3rd stage failed. The 3/8" chemical line fitting failed and did not hold the line in place. No injuries occurred during the event.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 23 Jan 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Valve (body, stem, plugs)

PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

Contract crew loosening 2" ball valve body caused 40 bbl scavenger spill into impermeable containment; no waters affected.

WHAT WENT WRONG?

On 1/22/2025, a contract crew was on location to remove used scavenger from the bubble tower, while in the process of removing a bull plug from a 2" ball valve so the vac truck could hook up and remove spent scavenger. The ball valve body started to unscrew instead of the bull plug, it started to leak slightly from the threads. Contractor continued to loosen the body not realizing there would be no way to stop the flow. This released 40 bbls of scavenger into impermeable containment. No waters of the US affected.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: 24 Jan 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Malfunctioning dump valve caused skim tank carryover spilling 30.43 bbl oil; area to be remediated.

WHAT WENT WRONG?

On 1/27/25, Production Technician discovered produced hydrocarbons had been released out of the thief hatch on the skim tank. After initial investigation it was determined that the dump valve malfunctioned forcing fluid to the skim tank at an increased rate resulting in the carryover. Area will be remediated.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Shutdown Systems - including operational well isolation and drilling well control equipment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 07 Feb 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Routine maintenance

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow

PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

Tank thief hatch released 18 bbl oil due to transmitter and pump malfunctions; all oil recovered in containment.

WHAT WENT WRONG?

On 2/5/2025 tank released 18 bbls oil into containment from the thief hatch. Level transmitters and salt water disposal pumps had both malfunctioned and issues were being actively worked at the time of the spill. Vac truck dispatched and recovered 18 bbls oil from containment. No injuries. No affected water ways. Incident investigation pending.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Shutdown Systems - including operational well isolation and drilling well control equipment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 18 Feb 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

6" PSV froze open releasing oil, water, and 2,628 KCF gas; spill occurred on private land and will be remediated.

WHAT WENT WRONG?

On 2/19/25, a spill occurred when the 6" PSV froze open due to inclement weather. The release occurred on private land. Area will be remediated.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

DATE: 20 Feb 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Valve (body, stem, plugs)

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Frozen 2" line valve broke releasing 134 bbl oil and 6 bbl water; partial recovery completed.

WHAT WENT WRONG?

On 2/19/2025 discovered a 2" bottoms line valve on the back of tank had frozen and broke releasing produced oil and produced water into earthen/permeable secondary containment. Free standing liquid removed. Remediation to resume 2/24/2025. No waters affected.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

DATE: 25 Feb 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Valve (body, stem, plugs)

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Frozen 3" load line valve ejected ball releasing 211 bbl oil; oil migrated slightly off pad; remediation ongoing.

WHAT WENT WRONG?

On 2/25/2025 discovered a 3" ball valve on the load line had frozen ejecting ball from valve and pushing line away from tank releasing 211 bbls of produced oil into permeable secondary containment. The produced oil soaked through the secondary containment and migrated a short distance off the pad to the south. The foreman, and environmental personnel were notified and mobilized to the site. Two vacuum trucks were used to remove oil from the secondary containment as well as off pad area. No waters, wildlife, or sensitive areas were affected. Remediation is ongoing.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

DATE: 03 Mar 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping material/tubing

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Small tornado damaged causing 113 KCF gas release from broken scrubber inlet; well shut in.

WHAT WENT WRONG?

On 3/4/2025 a Production Technician (PT) was notified by a transporter that a small tornado had just hit the well pad causing damage. The PT responded to the location within 30 minutes of the damage occurring. The scrubber to the gas lift had been knocked over and a 2" inlet broke free of the scrubber tower. The PT shut in the well and notified the foreman of the incident. No waters affected.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

DATE: 13 Mar 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline

PROCESS SAFETY FUNDAMENTAL: We recognize change.

INCIDENT DESCRIPTION:

Two corrosion failures on 10" transfer line caused total 838.29 bbl oil release; repairs and remediation underway.

WHAT WENT WRONG?

On 3/13/2025, environmental was notified of a spill that occurred on an underground 10" oil transfer line. While conducting the investigation for the 1st release a 2nd release on the same transfer line was reported approximately 1-mile northwest of the 1st spill area. The cause of these failures is due to internal corrosion. The 10" oil line was built/intended to be a crude oil line for 3rd party sales. The pipeline was not operated as a crude oil only line, but instead as an emulsion line. The 1st release happened on Company land, and the 2nd release happened on private land. Repairs will be completed by the pipeline/ops groups and environmental will handle the remediation.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 13 Apr 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping joint

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Gasket failure on 24" valve caused 4,800 KCF gas release; station evacuated and shut in remotely.

WHAT WENT WRONG?

On 4/14/2025 during pig launch operations crew experienced a gasket failure on a 24" ball valve. The line pressure at the time of the release was 1,046 PSIG. The impacted valve is located just downstream of the pig launch barrel. The personnel on site immediately evacuated the compressor station and the area was shut in remotely within five minutes. Initial release volume is unclear, but majority of unrecovered gas volume was released during blowdown at designated locations. Foreman, superintendent, environmental, and safety were all notified. Due to the remote nature of the location the public was not affected. Lease road was monitored for traffic and area was kept clear until it was safe to re-enter. Once blowdown was complete and site was gas tested personnel were allowed back in to inspect the area. Pipeline safety was notified and responded to the site. National Response Center notified. No waters affected.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Engineering to work with gasket manufacturer to confirm the correct material/gasket is being used for the correct application. Consider analysis of regional technology and gasket usage with other BUs.

Confirm torque specs for valve/gasket installation and frequency to re-torque.

Add ESD at facility entrance.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 14 Apr 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow

PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

Comms failure at tank battery caused 110 bbl PO release from water tank thief hatch; all product recovered.

WHAT WENT WRONG?

On 4/17/25, a comms failure happened on the battery. Around 3:30 am April 18th, Night automation tech was passing by tank battery and noticed oil coming out of the water tank thief hatch. Night tech equalized the tanks to prevent further spillage; transfer pumps were running at the time. Facility didn't send out alarm due to comm failure. No injuries occurred with this event. No affected water ways.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Implement the corrective action MOC at the Tank Battery to:

- Shutdown upstream ESPs due to the high tank levels at the Tank Battery
- Shutdown upstream ESPs during communication failures at the Tank Battery
- Open the anti-siphon valve based on the head switch high level
- Review other facilities for similar situation and implement MOC safeguards as necessary.

BARRIERS:

Hardware Barrier Failures: Shutdown Systems - including operational well isolation and drilling well control equipment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 06 May 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Equipment: Pump

PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

Pump housing failure from foreign object caused spill of 8.92 bbl oil and 2.97 bbl water; contained and recovered.

WHAT WENT WRONG?

On 5/5/25, environmental was notified of a spill from the circulating pump. The pump had a housing failure later found to be from a foreign object (steel bolt) entering the pump's fluid inlet. The failed pump was isolated quickly by personnel present at the central tank battery. The spill occurred on Company land. Area will be remediated

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 17 May 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)

PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

INCIDENT DESCRIPTION:

PSV lifted causing 43.485 Kscf gas release and light misting; liquids were minimal.

WHAT WENT WRONG?

On 05/11/2025, a release was discovered when excess pressure from wells lifted a PSV on a test separator, resulting in a gas release and light misting at the tank battery. The leak duration was approximately 7 minutes, and the total gas release was determined to be ~ 43.485 Kscf. Liquids were de minimis.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Shutdown Systems - including operational well isolation and drilling well control equipment

CAUSAL FACTORS:

NONE: Unspecified

DATE: 26 May 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping material/tubing

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Release of 16.7 bbl oil found during recycling; fluids remained onsite and recovered.

WHAT WENT WRONG?

On 5/28/2025, Lease Operator was recycling oil from tank, went back to site to check on the process and discovered oil pooled on the surface between the tanks and the treater. Process was shut down and free-standing fluid was recovered. No injuries occurred and all fluid remained on location. No US waters were affected.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 26 May 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Equipment: Pump

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Glycol pump failure released 483 kcf gas over 3.75 hours; site shut down and isolated.

WHAT WENT WRONG?

On 6/1/2025 Production Foreman received a low differential alarm. The foreman went to the Compressor site to investigate and discovered that the glycol pump on the reboiler skid had failed allowing gas to be released through a 3/4" port on the side of the pump. The site was shut down and isolated, proper notifications were made.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 26 May 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Equipment: Compressor/blower/fan

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Plant operator found 6" valve leaking causing 9.5 kcf natural gas release; area barricaded.

WHAT WENT WRONG?

At 7:30 A.M. CST a plant operator discovered a gas release coming from a compressor discharge 6" valve located inside the compression building. The operator notified their foreman and closed the valve. The valve was leaking by, so operations blocked off the area and established a barrier with caution tape. No injuries occurred and No US waters affected.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 27 Jun 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)

PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

Flare system ignited fluid after shutdown causing 9 bbl oil and 1 bbl water release; burned off onsite.

WHAT WENT WRONG?

On 6/26/25, Night Surveillance reported fluid carried over into the LP flare system and out the flare stack after the battery had a process shutdown. The fluid ignited and burned itself out on pad soil around base of flare. Investigation underway. No injuries occurred with this event. No affected water ways.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Protection Systems - including deluge and fire water systems

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 27 Jun 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping material/tubing

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Corrosion pinhole caused spill of 68.13 bbl oil and 158.97 bbl water; remediation to follow.

WHAT WENT WRONG?

Production Tech discovered that a pinhole had developed on the inlet header coming into the free water knock out due to corrosion. This caused a reportable spill to occur on the facility with fluid leaving location. Area will be remediated.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 09 Jul 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping joint

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Failed gasket on 8" flange caused 145 bbl oil spill; majority recovered; no water impacted.

WHAT WENT WRONG?

On 7/10/25 at 11:05 am, LO discovered a leak coming from an 8" flange due to a failed gasket. No injuries occurred with this event. No affected water ways.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 10 Jul 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Breaking Containment Locations: Loading/unloading coupling

PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

Truck loading error released 46.8 bbl oil; contained onsite and recovered.

WHAT WENT WRONG?

LO received a call from third-party trucking company stating they had a spill while loading truck. Driver forgot to open the valve to back section of the trailer causing overflow onto the ground outside the containment on permeable material. Standing fluid was collected and hauled to disposal. No injuries occurred and all fluid remained on location, no US waters were affected.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Emergency Response Equipment and Systems

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: 12 Aug 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping joint

PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

Power outage restart caused piping separation and fire at compressor station; no injuries.

WHAT WENT WRONG?

Following an overnight power outage, compressor station was being brought back online when process piping separated causing a fire. There were no injuries associated with the event.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

O2 Verification: Prior to restarting the station after a shutdown, verify oxygen levels with appropriate equipment to prevent combustion risks.

Vacuum Condition Check: Confirm with Operations that no vacuum conditions have occurred within the past 48 hours before restarting the compressor station.

Startup Safety Review: Before station startup, ensure that only essential personnel are onsite.

Low Point Monitoring: Monitor low-point pressure transmitter (PT) alarms on VRUs, VRTs, and tanks to detect potential vacuum conditions.

BARRIERS:

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: 16 Aug 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping joint

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Discharge line failure released 108 KCF gas; site blown down and isolated.

WHAT WENT WRONG?

On 8.20.25 a discharge line from the gas lift compressor failed causing segments of the discharge line to separate and eject up to 150 feet from the compressor area. An employee was notified by call-out that the gas lift compressor had automatically shut down. Upon arriving to the location, the employee observed a release coming from the dump line off of the second and third stage scrubbers. All wells were shut in and blown down. No waters were affected.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Consider integrating air compressors (suction controller supply) into the main ESD system at the facility entrance. This would allow simultaneous shutdown of compressors and wells during emergencies. Evaluate implications for wells using plungers or electric actuated valves.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 26 Aug 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping material/tubing

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Internal corrosion on tester oil leg caused 21.5 bbl spill; remediation planned.

WHAT WENT WRONG?

Production Tech received a dispatch from the remote operations hub about a leak. When the production tech arrived, he found the oil leg from a tester was leaking due to internal corrosion. This caused a reportable spill to occur on the facility with fluid leaving location. Area will be remediated.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 16 Sep 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Instrumentation and small bore tubing

PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

INCIDENT DESCRIPTION:

Nipple failure on third stage PSV caused 320 KCF gas release; operator shut down on discovery.

WHAT WENT WRONG?

On 9/18/2025, a 1" double XX nipple on the third stage PSV and Hot Gas discharge line of a compressor failed. At 11:15 PM, an oil hauler arrived on location and noticed a loud noise coming from the compressor area. The oil hauler left the location, marked his ticket as "No Load," and contacted his dispatch supervisor. The dispatch supervisor attempted to notify the contract lease operator via call and text and text to Field Foreman. No responses were received during the night. At 6:00 AM, the LO was dispatched to location and upon arrival, the operator confirmed the nipple failure and immediately closed the suction valve and activated the compressor ESD. No waters affected.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Continue engineering efforts for the updated design change to eliminate vibration-prone configurations. Recognize and commend the driver who demonstrated safety leadership by stopping work and evacuating. Verified shutdown setpoints on other compressors

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 23 Sep 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Equipment: Pump

PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

Seal failure caused 180 bbl oil release; fully recovered in containment.

WHAT WENT WRONG?

On 9/30/25, Night lease operator discovered the triplex pump seals failed. No injuries occurred with this event. No affected water ways.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Maintenance Procedure requirements are not well known by internal or external maintenance personnel and therefore were not followed for packing gland nut installation/adjustment.

BARRIERS:

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: 10 Oct 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Gas buster controller failure released 20 bbl oil; fully recovered.

WHAT WENT WRONG?

On 10/25/25, LO discovered the gas buster liquid level controller malfunctioned releasing gas to the battery causing the pressure relief valve to lift. No injuries occurred with this event. No affected water ways.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Shutdown Systems - including operational well isolation and drilling well control equipment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 01 Dec 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Instrumentation and small bore tubing

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Pressure transmitter failure released 355 bbl oil; 250 bbl recovered.

WHAT WENT WRONG?

On 12/02/25, LO discovered and 3/4" pressure transmitter on the LACT unit had failed releasing fluid. No injuries occurred with this event. No affected water ways.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 14 Dec 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Valve (body, stem, plugs)

PROCESS SAFETY FUNDAMENTAL: We walk the line.

INCIDENT DESCRIPTION:

2" valve left open released 2.29M lbs gas; operator closed valve; no waters affected.

WHAT WENT WRONG?

On 12/18/25 a third party notified Company of leak, when operator arrived a 2" valve was open releasing gas. Operator closed the valve stopping the leak. No U.S. waters affected.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

DATE: 18 Dec 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Instrumentation and small bore tubing

PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

INCIDENT DESCRIPTION:

Vibration-related nipple failure released 23 bbl oil; partial recovery completed.

WHAT WENT WRONG?

On 12/19/25, LO discovered nipple connecting to discharge PT failed to vibration from DC line. No injuries occurred with this event. No affected water ways.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 18 Dec 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Instrumentation and small bore tubing

PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

INCIDENT DESCRIPTION:

Vibration failure on lease automatic custody transfer 130 discharge pressure transmitter caused 80 bbl oil spill; 60 bbl recovered.

WHAT WENT WRONG?

On 12/21/25, lease operator discovered discharge pressure transmitter failed off of discharge pipe due to vibration. No injuries occurred with this event. No affected water ways.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 18 Dec 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Instrumentation and small bore tubing

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Glycol pump nipple failure released ~1 gallon glycol and 137.8 kscf gas; fluid collected for disposal.

WHAT WENT WRONG?

There was a loss of approximately 1 gallon of glycol into containment and 137.8 kscf/5992lbs of gas to the atmosphere. The source of the release was a faulty pipe nipple on the glycol pump for the dehydrator. The contaminated fluid from containment will be pumped into the on site storage tank for proper disposal. Waters were not affected.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 18 Dec 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Breaking Containment Locations: Breaking containment location

PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

Bottom valve sand blockage masked fluid level, releasing 43 bbl oil during manway opening; no injuries.

WHAT WENT WRONG?

On 12/19/25, Maintenance crew was tasked with cleaning out Bulk tank. Vessel was isolated correctly from process and was in the process of removing sand from a bottom valve. After beginning to get air through the hose the crew lanced the valve opening to confirm no sand blockages in addition to confirm no levels on the site glass. With no returns the crew proceeded to close the belly valve and commenced opening the manway. Following removal of all manway bolts with the exception of the bottom bolts the crew confirmed no fluid present and proceeded to remove the final bolts and open the manway. While opening the manway fluid began to release behind sand build up. Other bottom vessel openings for cleaning flanged without isolation valves. No injuries occurred with this event. No affected water ways.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

DATE: 20 Jan 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Instrumentation and small bore tubing

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Failure of 1/2 inch nipple on PSV line released 26,279 lbs CO₂ inside compressor building; safety systems worked and unit shut down.

WHAT WENT WRONG?

During normal operations, the failure of a 1/2 inch nipple on the second stage discharge PSV line allowed for 26,279 lbs of CO₂ to be released inside the compressor building. The facilities atmospheric safety systems operated properly alerting operations. The compressor shutdown and the building ventilation systems engaged.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 24 Feb 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Equipment: Filter

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Sour water pre-filter O-ring leak triggered H₂S alarms; facility isolated, ventilated, and area reclassified from air-pack available to carry zone.

WHAT WENT WRONG?

On 2/24/2025, the sour water pre-filter leaked through the O-ring. Three H₂S detectors alarmed in the building.

Console operator closed shut down valve on A slug catcher to stop sour water flow to Sour Water Pump House (SWPH). Once concentration dissipated, outside operator isolated filter.

Outside operators did not enter SWPH until concentration indicated below IDLH. Outside operators opened doors to ventilate, then isolated filter.

SWPH was deemed an air pack “available” area prior to this incident and has since been deemed a “carry” area.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 03 Mar 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Equipment: Compressor/blower/fan

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Head gasket failure on first stage cylinder released 2,948 lbs CO₂; no hazardous atmosphere detected and no personnel affected.

WHAT WENT WRONG?

Operations identified head gasket failure on compressor first stage cylinder after variable pockets were installed and being monitored T restart. 2,948 lbs of CO₂ released. No fixed monitoring or personnel gas detectors registered a hazardous atmosphere during the event. No personnel impacted and no other equipment damaged occurred.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 04 Apr 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Valve (body, stem, plugs)

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Gasket failure on Test Treater #1 released 17.6 bbl oil inside building and 4.5 bbl outside; all contained and recovered.

WHAT WENT WRONG?

A crude oil release occurred due to a gasket failure on Test Treater. An estimated 17.6 barrels of oil accumulated within the treater building, and 4.5 barrels migrated outside of the building. All fluids remained within secondary containment and have been recovered via vacuum truck.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 21 Apr 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping joint

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Operator heard leak sound and slight H2S odor in compressor building; evacuated and compressor shut down; trending later showed 2.6 ppm.

WHAT WENT WRONG?

While making rounds, the operator entered the compressor building and heard what sounded like a leak and smelled a slight odour of H2S. He immediately evacuated the building and notified the foreman to shut down the compressor. Neither personal nor fixed HS monitors alarmed at the time but looking at the trending of the building monitors, it was noted that they saw 2.6 ppm.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 21 Apr 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Valve (body, stem, plugs)

PROCESS SAFETY FUNDAMENTAL: We maintain safe isolation.

INCIDENT DESCRIPTION:

Recycle pump suction valve and check valve failed, releasing 0.31 bbl oil inside building and 17.8 bbl outside; contained within facility.

WHAT WENT WRONG?

The recycle pump suction valve was closed by appearance, but failed to seal completely. The suction check valve to the pump also failed, resulting in a release of crude oil through the recycle sump inside of the building. A total of

0.31 barrels crude oil collected inside of the building, plus 0.23 bbl sludge/solids. A total of 17.8 barrels crude oil migrated outside of the building.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 12 May 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping joint

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Foreman found small head gasket leak on HP compressor #4 cylinder; compressor shut down; trending later showed 1.2 ppm.

WHAT WENT WRONG?

The foreman was driving past the open doors of the compressor building and heard what sounded like a leak. He stopped to investigate and found a small head gasket leak on the #4 cylinder of the HP machine. He immediately shut down the compressor. The operator and M&R mechanic had been through the compressor building on rounds just 30 minutes prior and noticed no leak. Neither personal nor fixed H2S monitors alarmed at the time but looking at the trending of the building fixed monitors, it was noted that they saw 1.2 PPM.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 30 Jun 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Fiberglass flowline failure released 7,067.35 lbs CO2 in ROW; no off-site impact.

WHAT WENT WRONG?

The plant operator received a CO₂ high injection flowrate alarm for the injection well. After dispatching the field operator, it was found that the injection well fiberglass flowline had failed, releasing 7,067.35 lbs of CO₂ in the ROW. There was no off-site impact.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 08 Aug 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping material/tubing

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

1-inch equalization line leak released CO₂/H₂S at 320 lb/hr (14.5 lb/hr H₂S).

WHAT WENT WRONG?

On 8/8/2025, gas detection alarmed on 10ppm H₂S when a 1" equalization line in the mezzanine leaked downward (~1050 psig / 5% H₂S). Train was promptly isolated and depressured to stop the event. The maximum concentration at grade was 22ppm approximately 10 minutes into the event. The leak size was measured to be 1/16". Total Leak Rate = 320 lb/hr of which 14.5 lbs/hr was H₂S, which exceeds Tier 1 Thresholds indoors. This is the peak rate of leakage, however, the event went on for around 10 minutes (1.45 lbs/hrs) followed by depressuring of the unit over the next hour.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 28 Oct 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Fiberglass flowline failed releasing 6,141 lbs CO₂ in ROW; Tier 1 event; no off-site impact.

WHAT WENT WRONG?

A landowner notified operations of a possible CO2 leak. Upon investigation, the operator discovered the fiberglass flowline had failed. This caused 6141 lbs of CO2 to be released on the ROW. There was no off-site impact. This incident is classified as a Tier 1.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 07 Dec 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Routine maintenance

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank

PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

Water tank overpressure launched tank 250 ft and released 290 bbl water inside berms; investigation ongoing.

WHAT WENT WRONG?

The operator 1.8 miles away was leaving another facility when he noticed white steam near the Section 30 Tank Battery (TB). Upon arriving at the Section 30 TB, the operator identified the 500 bbl water tank had experienced an overpressure event. The tank lifted from the TB, landing 250' to the east at the adjacent well pad. No personnel were in the area at the time. The 290 bbls of produced water in the tank were released to the facility and remained inside of the containment berms. The well experienced minor damage to the 1" and 2" valve on the wellhead. A minor volume of emulsion collected in the cellar.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 08 Dec 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Breaking Containment Locations: Loading/unloading coupling

PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

Production tank load line 3-inch valve failed releasing 228.56 bbl crude into berms; operator isolated line; fluids remained contained.

WHAT WENT WRONG?

At 5:34 am, the operator received a dike alarm which shut in the facility. Upon arrival at the site they identified that the production tank oil load line 3" valve had failed, releasing 228.56 bbl crude oil into the site berms. All fluids remained inside the containment berms. The operator isolated the load line via an upstream valve.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 17 Dec 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Equipment: Compressor/blower/fan

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Pressure gauge failure compressor released ~3,180 lb/hr CO₂; detected during operations; no monitors alarmed.

WHAT WENT WRONG?

A CO₂ release occurred due to the failure of a pressure gauge on the 2320 compressor. An initial calculated volume of 3,180 lb/hr was released to the atmosphere. The release was discovered by a mechanic during normal operations. No personal or fixed monitors alarmed. This event is under investigation.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 20 Jan 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Breaking Containment Locations: Piping/valve (inadvertently left) open to atmosphere

PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

The operator left pump bleed valve open after failed priming, causing sump overflow contained in unlined berm.

WHAT WENT WRONG?

The operator arrived on location and found that his circulation pump sump was running over. The operator had tried to prime pump the day before and after unable to prime he shut off the pump at the panel box but did not shut the bleed valve on the pump which caused the spill. Spill was contained inside the unlined berm. All notifications were made.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

DATE: 04 Feb 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Valve (body, stem, plugs)

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Failed suction-side valve filled sump and spilled 84 bbl oil inside unlined containment; 76 bbl recovered.

WHAT WENT WRONG?

The operator discovered the 1" valve on the suction side of the circulating pump failed, filling up the sump and spilling 84bbls of oil on the ground inside of the tank battery unlined containment. Recovered 76bbls of oil. Habitat not affected.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 27 Mar 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow

PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

Power/coms loss caused tanks to run over, releasing oil/water inside firewall and small amount on pad; recovery done.

WHAT WENT WRONG?

Production tanks ran over after a power failure and communications loss lasting 13 hours. This allowed the release of 171 barrels of oil and 705 barrels of produced water contained inside the lined firewall. 1 barrel of oil and 25 barrels of produced water spilled onto the facility pad. 162 barrels of oil were recovered from inside the firewall and 0.5 barrels was recovered from the pad. 670 barrels of produced water were recovered from inside the firewall and 23 barrels were recovered from the pad. Technician will verify tank sticks are functioning correctly and environmental will assign the remediation to a contractor. The superintendent and environmental were notified.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Shutdown Systems - including operational well isolation and drilling well control equipment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 04 Apr 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow

PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

SDV failed to close causing spill in unlined containment; inlet valve shut and recovery initiated.

WHAT WENT WRONG?

Technician had arrived to location when spill was discovered. Spill was caused by failure of the SDV not closing. Inlet valve was shut to stop spill, and trucks were dispatched to start recovering fluids on the ground. Spill was released inside the unlined containment and the facility. Site will remain shut in until all repairs are made.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Shutdown Systems - including operational well isolation and drilling well control equipment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 14 Apr 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Tank leaked due to internal corrosion, spilling 111 bbl. oil inside unlined berm; 17 bbl. recovered.

WHAT WENT WRONG?

Environmental crew working on location found tank leaking on Tank, on bottom side of tank spilling 111 bbl. oil all inside unlined berm, vac truck called out transfer oil to off tank. And picked up 17 bbl. oil inside berm. Lost 94 bbl. oil to ground. Due to internal corrosion. Notified superintendent, safety, and environmental.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 04 Jun 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow

PROCESS SAFETY FUNDAMENTAL: We stop if the unexpected occurs.

INCIDENT DESCRIPTION:

Power loss caused inlet valve to fail open and tanks overflowed; manual valve closed until power restored.

WHAT WENT WRONG?

The Technician arrived to location and noticed the tanks overflowing. It was determined that the facility had lost power the night before causing the inlet valve to fail open. The production batteries had the power restored and the facility stayed without power. The Tech closed the manual valve on the inlet until the power was restored.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Shutdown Systems - including operational well isolation and drilling well control equipment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 03 Jul 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow

PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

Coms loss caused skim tanks to overflow; tank sticks, transmitter, and pressure switches replaced.

WHAT WENT WRONG?

Due to loss of coms, Tech went to location to verify tank levels. Upon arrival, tech noticed skim tanks overflowing. Inlet did not shut in on skim tanks HiHi level. Tech reached out to maintenance to get parts replaced and tested. Upon investigation, tank sticks on two tanks were reporting 0 when the tanks were full. Tank sticks and outdated transmitter were replaced. Pressure switches were also replaced, because they could not be recalibrated to the correct level. Transducer was cleaned and returned to normal operation.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Shutdown Systems - including operational well isolation and drilling well control equipment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 31 Jul 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow

PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

PLC programming loss prevented alarms/shut-ins, causing overflow; inlet shut manually and remediation started.

WHAT WENT WRONG?

During normal operations contractor onsite experienced coms loss causing the shutdown of their equipment. Vendor assumed inlet skid had shut in as well but due to loss of programming in PLC it did not. Tanks continued to fill and Hi and HiHi level alarms were not triggered in PLC. When contractor saw overflow inlet was shut manually and remediation began. Vac trucks called to recover what was possible in lined containment and facility. Programming reinstalled and tested before facility brought back online.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Shutdown Systems - including operational well isolation and drilling well control equipment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

DATE: 20 Sep 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow

PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

Check valve failure on transfer pumps caused thief hatch releases and overfilled three water tanks.

WHAT WENT WRONG?

Operator found fluid being release from the thief hatches due to a check valve failure on the water transfer pumps. Tank sticks were not in proper operation, and we ran over 3 water tanks.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 25 Sep 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Lightning strike caused salt water disposal unit (SWD) tank fire, burning 18.75 bbl. oil and damaging multiple tanks before extinguished.

WHAT WENT WRONG?

The Operator saw smoke in field after a storm passed and discovered the SWD had been struck by lightning. SWD was being used only for fresh water remediation for the water wells. Tanks on location were (1) 300 bbl. steel tank used as a skim tank containing 1' of oil and 2'6" of water. 18.75 bbl. of oil was burned off in the fire. (3) 500 bbl. fiberglass tanks containing a combined 500 bbl. of fresh water and (1) 210 bbl. open top fiberglass tank (empty) were lost to the fire. Fire department was dispatched an fire was extinguished. Area around the facility on the west and south side show signs of fire damage.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Ignition Control

CAUSAL FACTORS:

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

DATE: 03 Oct 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Unknown/insufficient information: Unknown/insufficient information

PROCESS SAFETY FUNDAMENTAL: Unspecified.

INCIDENT DESCRIPTION:

Fire on Oct 3, 2025; wells shut in and fire extinguished without injuries.

WHAT WENT WRONG?

On October 3rd, 2025 a fire erupted at a tank battery. The fire was initially detected at 1:50pm prompting immediate shut-in of all wells associated with the site. Emergency crews arrived on site by 2:20pm and fire suppression efforts were in place. The fire was ultimately extinguished with no injuries to report.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Ignition Control

CAUSAL FACTORS:

NONE: Unspecified

DATE: 09 Oct 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Wells, drilling and intervention: Well

PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

Oxidizer/acid job miscommunication led to overpressure event ejecting wellhead; procedural gaps documented.

WHAT WENT WRONG?

While attempting to acidize a saltwater disposal well, an unconfirmed downhole chemical reaction resulted in an overpressure event that separated the wellhead from the casing strings, resulting in the wellhead being ejected approximately 20'.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

No procedure, No SPAC, No preparation, No supervision, Enforcement NI.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: 02 Dec 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Instrumentation and small bore tubing

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Pipeline Lease Automatic Custody Transfer (LACT) unit leak from nipple under flow meter spilled 33 bbl. on pad; 30 bbl. recovered.

WHAT WENT WRONG?

Night operations was doing final rounds and came upon spill coming from pipeline lact. After investigating, night PO observed leak coming from nipple underneath flow meter. Immediate action, isolated lact and notified foreman on-call. Caused 33BBL spill on pad, vac truck recovered 30BBL's. All fluid on pad, no harm to habitat.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 12 Dec 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline

PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

Contractor vehicle hit gas injection blowdown riser releasing gas; compressor shut in and no injuries.

WHAT WENT WRONG?

Contractor backed over a high-pressure gas injection blowdown riser with company vehicle causing a gas release. Compressor shut in due to low discharge pressure. No injuries reported, all notifications were made. On December 12, 2025, two contract chemical technicians (CT) arrived at a well pad to install equipment for methanol injection. After reviewing the job scope and determining what additional parts were required, CT-1 left to retrieve the parts, while CT-2 remained on site to begin laying out materials for installation. As CT-1 was exiting the location, they realized parts for CT-2 had not been left. While reversing in a straight line from the entrance/exit with the intention to turn right and return to CT-2, CT-1 struck a high-pressure gas injection blowdown riser (~900 psi) located near the entrance. The impact damaged the line, causing a gas release of approximately 18,300 SCF / 1170 lbs. The compressor automatically shut down due to low discharge pressure. No injuries occurred, and all required notifications were completed promptly.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

DATE: 25 Feb 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Other production

POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)

PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

During hot water flush, well started unloading, and the PSV released on the LP separator.

WHAT WENT WRONG?

Incorrect pressure settings.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

DATE: 16 Mar 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Breaking Containment Locations: Loading/unloading coupling

PROCESS SAFETY FUNDAMENTAL: We walk the line.

INCIDENT DESCRIPTION:

Oil hauler failed to check his truck before loading and spilled.

WHAT WENT WRONG?

Human performance.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

DATE: 28 Mar 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Upset

POINT OF RELEASE: Equipment: Pump

PROCESS SAFETY FUNDAMENTAL: We recognize change.

INCIDENT DESCRIPTION:

Oil released from a pump.

WHAT WENT WRONG?

Pump failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 16 Apr 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Upset

POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)

PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

Cooler pressure safety valve released.

WHAT WENT WRONG?

Premature pressure relief.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 25 Apr 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Other production

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Valve (body, stem, plugs)

PROCESS SAFETY FUNDAMENTAL: We walk the line.

INCIDENT DESCRIPTION:

During batch treatment on the flowline, a gas lift line was misidentified as the flowline. A bull plug was misidentified and opened.

WHAT WENT WRONG?

Human performance.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

DATE: 24 May 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Upon arriving for rounds, operator discovered that the flowline had developed a leak underground and was flowing to the surface.

WHAT WENT WRONG?

Corrosion.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

DATE: 17 Jun 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Relief, Vent and Discharge Systems: Drain

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Drain valve on air eliminator opened, and spill occurred.

WHAT WENT WRONG?

Valve failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

DATE: 29 Jun 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Relief, Vent and Discharge Systems: Drain

PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

Drain valve on air eliminator opened, and spill occurred.

WHAT WENT WRONG?

Human performance.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

DATE: 11 Jul 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Other production

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank

PROCESS SAFETY FUNDAMENTAL: We control ignition sources.

INCIDENT DESCRIPTION:

Lightning strike on tanks, and fire followed.

WHAT WENT WRONG?

Lightning.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Ignition Control

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

DATE: 16 Jul 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow

PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

Tanks overflow due to level switch error.

WHAT WENT WRONG?

Automation error.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

DATE: 24 Jul 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Other production

POINT OF RELEASE: Relief, Vent and Discharge Systems: Relief valve (body, plugs)

PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

Pressure safety valve released on low pressure sweep line.

WHAT WENT WRONG?

Incorrect valve setup.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

DATE: 01 Aug 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Planned shutdown

POINT OF RELEASE: Equipment: Pressure vessel

PROCESS SAFETY FUNDAMENTAL: We control ignition sources.

INCIDENT DESCRIPTION:

A localized flash fire occurred inside the injection compressor's 1st stage knockout (KO) drum within a central facility. Personnel had entered the drum to clean and demolish internal components in preparation for system upgrades. Non-intrinsically safe lighting and tools had been permitted under a hot work permit along with appropriate confined space entry permit. Flammable vapors ignited resulting in burns.

WHAT WENT WRONG?

Confined space mismanagement.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

DATE: 19 Aug 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

A hole developed on the bottom side of the slop tank.

WHAT WENT WRONG?

Corrosion.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 24 Sep 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Planned shutdown

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Valve (body, stem, plugs)

PROCESS SAFETY FUNDAMENTAL: We walk the line.

INCIDENT DESCRIPTION:

Recirculation pump inlet valve was left open.

WHAT WENT WRONG?

Incorrect valve setup.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

DATE: 06 Oct 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Workover/Well services

MODE OF OPERATION: Other production

POINT OF RELEASE: Wells, drilling and intervention: Well intervention equipment

PROCESS SAFETY FUNDAMENTAL: We walk the line.

INCIDENT DESCRIPTION:

Tubing head penetrator failed, resulting in a release.

WHAT WENT WRONG?

Installation error.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

DATE: 13 Oct 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Routine maintenance

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping material/tubing

PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

Discharge line that had a 1" riser with a 1" ball valve was accidentally run over. Upon striking the riser, it broke off, and gas began to release.

WHAT WENT WRONG?

Vehicle accident.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

DATE: 14 Oct 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Sight glass

PROCESS SAFETY FUNDAMENTAL: Unspecified.

INCIDENT DESCRIPTION:

Test treater sight glass broke.

WHAT WENT WRONG?

Sightglass break.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: 03 Nov 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping joint

PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

INCIDENT DESCRIPTION:

Valve nipple backed out of threads due to vibration on piping.

WHAT WENT WRONG?

Valve failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

DATE: 03 Dec 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Planned shutdown

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank

PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

Water tank ruptured on the top due to overpressure from a mechanical fault on a water control valve on production separator.

WHAT WENT WRONG?

Tank overpressure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

DATE: 06 Dec 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping joint

PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

While releasing pressure on a pipe, the pipe rotated and contacted with person, causing injury.

WHAT WENT WRONG?

Unexpected rotation.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

DATE: 12 Dec 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Equipment: Pump

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

The circulation pump seal failed resulting in a spill.

WHAT WENT WRONG?

Seal failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 29 Dec 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Other production

POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)

PROCESS SAFETY FUNDAMENTAL: We recognize change.

INCIDENT DESCRIPTION:

Cooler pressure safety valve released.

WHAT WENT WRONG?

Pressure relief design issue.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Human Barrier Failures: Acceptance of handover or restart of facilities or equipment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

DATE: 15 Feb 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Relief, Vent and Discharge Systems: Relief valve (body, plugs)

PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

INCIDENT DESCRIPTION:

Incident 1: A threaded test port for a pressure relief valve on a compressor system failed due to fatigue induced by vibration.

WHAT WENT WRONG?

Fatigue induced vibration.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Issue was not addressed after previous failures, vibration preventative maintenance program was not systematic.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 21 Feb 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Relief, Vent and Discharge Systems: Relief valve (body, plugs)

PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

INCIDENT DESCRIPTION:

Incident 2: A threaded test port for a pressure relief valve on a compressor system failed due to fatigue induced by vibration.

WHAT WENT WRONG?

Fatigue induced vibration.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Issue was not addressed after previous failures, vibration preventative maintenance program was not systematic.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 15 Mar 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Breaking Containment Locations: Piping/valve (inadvertently left) open to atmosphere

PROCESS SAFETY FUNDAMENTAL: We walk the line.

INCIDENT DESCRIPTION:

During seal replacement on a condensate pump, a valve was left open post-maintenance, leading to condensate release.

WHAT WENT WRONG?

Following maintenance on a condensate pump a bleed valve left open, allowing release of condensate to the skid.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

LOTO procedure incomplete, activity completed that was not included in the CoW permit scope, alarm acknowledged but not responded to.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)

DATE: 15 Aug 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Start-up

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Choke

PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

INCIDENT DESCRIPTION:

After bringing the well online following a frac hit, there was a release downstream of the 90-degree choke on the high-pressure separator.

WHAT WENT WRONG?

Washout occurred in the dump system adjustable choke due to redirected flow from the sand damaged choke stem.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Unknown integrity issues in the equipment from the previous operator, with existing wall loss/ erosion and flow of solids through the equipment.

BARRIERS:

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Acceptance of handover or restart of facilities or equipment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

DATE: 19 Sep 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Equipment: Compressor/blower/fan

PROCESS SAFETY FUNDAMENTAL: We recognize change.

INCIDENT DESCRIPTION:

Cylinder valve cover on a compressor was ejected off allowing a release of gas through the open cylinder.

WHAT WENT WRONG?

The securing stud that held the valve assembly in place snapped, causing the valve's front component and plate to dislodge. The dislodged valve components hammered against the valve cover, causing the valve cover to fail and be ejected off the cylinder.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Valve caps inadequately torqued.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 07 Jan 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Start-up

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping joint

PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

While bringing a compressor back online, a fire started at the front of the compressor.

WHAT WENT WRONG?

Valve not tightened properly.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: 08 Jan 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Other production

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline

PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

While trenching to lay an electrical line, equipment contacted a high-pressure gas injection line.

WHAT WENT WRONG?

Complete line marking.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: 18 Jan 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Other production

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Leak found on pipeline during operator visit.

WHAT WENT WRONG?

Corrosion.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 27 Jan 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Equipment: Fired heater/Boiler/Furnace

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Two heaters caught fire and sustained damage.

WHAT WENT WRONG?

Gasket failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 20 Aug 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Equipment: Pump

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

41.312 crude oil released, 32 bbls being in containment and 8.7 out. It happened because the seal on transfer pump blew out, right after a new one was installed on the same day as spill happened.

WHAT WENT WRONG?

Transfer pump blew out.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Unknown.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

NONE: Unspecified

DATE: 24 Aug 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Unspecified - other

MODE OF OPERATION: Normal

POINT OF RELEASE: Equipment: Pump

PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

A fire incident occurred at the Central Tank Battery (CTB), with preliminary assessments indicating that the origin point was Lease Automatic Custody Transfer (LACT) unit Pump B. The ignition source remains undetermined at this time. The fire resulted in extensive damage to The LACT system, adjacent storage tanks, and MCC building. The estimated product involved with the incident is approximately 182 barrels of crude oil.

WHAT WENT WRONG?

LACT Pump failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Inconsistent workflows on performing work, Miscommunication of severity of pump damage, and better procedure for taking equipment out of service.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

NONE: Unspecified

NORTH AMERICA OFFSHORE

DATE: 27 Jul 2025

COUNTRY: Canada

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Equipment: Pump

PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

During an offshore cargo offloading operation, a leak was identified on a cargo pump and offloading was suspended. The pump was shut down, the receiving vessel was disconnected, and response actions were taken; an estimated volume of oil was released from the pump area.

WHAT WENT WRONG?

A steam fitting on cargo pump No. 1 failed, resulting in loss of containment. The investigation notes the fittings remained in place after their original use was no longer required, and the fitting was not specifically included or tested within routine PM scope.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Remove or positively isolate legacy connections that are no longer required, and ensure PM/inspection scopes explicitly include remaining fittings and potential leak points. Confirm operating conditions that can introduce vibration or abnormal loads are evaluated as part of readiness reviews prior to offloading.

BARRIERS:

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Surveillance, operator rounds and routine inspection

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: 05 Mar 2025

COUNTRY: Mexico

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping joint

PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

During a walkthrough on the platform, a worker who was holding the position of Helper C perceived an abnormal noise near the fuel gas regulation package, which operates at 74 kg/cm². Upon inspecting the package, he identified the presence of gas in a 2" Ø x 600# line located on the mezzanine of the first level.

As he inspected the line more closely, he detected a loss of containment at the threaded connection between the 2" Ø x 600# line and the 2" x 600# coupling. He proceeded to tighten the 2" Ø coupling to the 2" Ø pipe, observing that the gas emission gradually decreased. Immediately afterward, the threaded coupling detached, leaving his left palm exposed to the gas coming out of the pipe.

He immediately informed the Control Room and secured the residual gas system. With support from the first aid brigade of the Processing Centre, he was transferred, conscious and walking on his own, from platform to the Processing Centre via lateral flight for medical attention.

WHAT WENT WRONG?

A loss of containment occurred at a threaded connection on a high-pressure fuel gas line (~74 kg/cm²).

The worker attempted to tighten a leaking threaded coupling while the line was pressurized, which is unsafe.

Tightening the live connection caused the threaded coupling to fail and detach completely, exposing the worker's hand directly to escaping gas.

Threaded fittings are not robust for high-pressure gas service and are prone to failure due to vibration, wear, misalignment, or improper makeup.

Immediate isolation and depressurization were not performed before intervention, increasing the risk of injury and escalation.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Never tighten or repair a leaking pressurized gas connection. Any gas leak must trigger stop work, evacuation, notification to the Control Room, isolation, and depressurization.

Threaded connections should be avoided in high-pressure fuel gas systems; welded or flanged joints are safer and more reliable.

Workers must recognize “line of fire” and stored energy hazards and apply Stop-Work Authority without hesitation. Clear procedures and training are essential for responding to gas leaks, emphasizing that leak mitigation is not a field adjustment task.

Preventive maintenance and inspections should focus on known weak points such as threaded joints, especially in vibrating areas like mezzanines.

Early detection by the worker and prompt communication with the Control Room helped prevent a more severe outcome, highlighting the value of situational awareness and reporting.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 28 Apr 2025

COUNTRY: Mexico

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Subsea: Subsea pipeline/flowline

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

The subsea support vessel reported that during ballast removal activities on the Line, diving personnel identified, during the 06:00 technical schedule, a loss of containment located 57 cm from the field joint.

Subsequently, while continuing cleaning activities for resizing, a second loss of containment was identified 25 cm from the field joint, also in the direction of platform. Personnel from the platform carried out operational adjustments to reduce the operating pressure of the outlet pipeline from the platform to the terminal from 23 to 20 kg/cm². The quantification of the iridescence area and volume is in progress, with environmental impact yet to be determined.

WHAT WENT WRONG?

During ballast removal and cleaning activities, two separate losses of containment were identified on the same pipeline:

- First leak 57 cm from the field joint.
- Second leak 25 cm from the field joint, toward the platform.

The discovery of a second leak while cleaning indicates degraded pipeline integrity, possibly due to corrosion, coating failure, or mechanical damage near the field joint area.

The line continued operating until pressure was reduced from 23 to 20 kg/cm², rather than being fully isolated.

The presence of iridescence confirms a hydrocarbon release, with environmental impact still under evaluation.

The proximity of both leaks to the field joint suggests a systemic weakness, not an isolated defect.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Field joints are critical integrity points and require enhanced inspection, monitoring, and protection, especially during subsea or underwater activities.

Cleaning and resizing activities can expose or worsen existing defects; therefore, integrity verification should be performed before and during such work.

When multiple leaks are detected, pressure reduction alone may be insufficient; isolation and shutdown criteria must be clearly defined and applied.

Early detection by diving personnel highlights the importance of qualified inspections during subsea operations. Environmental response readiness must be immediate, including rapid quantification of release and impact assessment.

Findings should trigger a broader integrity assessment of similar lines and joints to prevent recurrence.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 12 Aug 2025

COUNTRY: Mexico

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping material/tubing

PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

The worker, holding the position of Plant Specialist Operator, was splashed with hot liquid (amine). He was transported via regular flight to the hospital for evaluation by a specialist. His health condition is stable, and one day of medical leave was issued.

WHAT WENT WRONG?

A Plant Specialist Operator was splashed with hot amine, indicating a loss of control of a hazardous chemical during plant operations.

The splash suggests inadequate containment, shielding, or distance from the hot amine source at the moment of the task.

Possible contributing factors include equipment leakage, improper handling, insufficient barriers, or procedural gaps when working with hot fluids.

Although the injury was not severe, the event required hospital evaluation and medical leave, confirming it was a recordable exposure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Hot amine systems present chemical and thermal burn hazards and must be treated as high-risk during operation and intervention.

Engineering controls (splash guards, insulation, shields, proper drainage) are critical to prevent direct exposure.

Strict adherence to PPE requirements (chemical-resistant gloves, face shield, goggles, flame-resistant clothing) is essential when working near hot amine.

Procedures and job safety analyses (JSA) should clearly identify splash and burn hazards and define safe positioning and distancing.

Even minor exposures highlight the need for continuous training on chemical hazard awareness and line-of-fire control.

Prompt medical response and transportation demonstrate good emergency response, but prevention remains the primary objective.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

DATE: 13 Mar 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Workover/Well services

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping material/tubing

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

During a commissioning pressure test on newly installed equipment, the system failed to hold pressure. A field inspection identified a small leak from a tubing union fitting below a platform. The associated pump was shut down and the system pressure was safely relieved.

WHAT WENT WRONG?

A tubing leak occurred in the turret resulting in the release of MeOH in the turret area.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Prior to the incident it was believed that all significant hydrocarbon and MeOH leaks in the turret could be detected without human response/intervention. The incident revealed that this is not correct and that protocols and instructions have to address the aspect of hydrocarbon presence in the turret area that is to be considered as a confined space.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 20 Aug 2025

COUNTRY: USA

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping joint

PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

During late-afternoon operations, a small-diameter tubing connection failed resulting in a nitrogen release from a high-pressure nitrogen line. A nearby worker was startled by the sudden release and injured his shoulder while moving away from the area.

WHAT WENT WRONG?

Nitrogen (N₂) tubing connection failed while workers were working in the area.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

The worker making up the fitting may have applied insufficient force or applied too much force to the fitting for the fitting to perform as designed.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

RUSSIA & CENTRAL ASIA ONSHORE

DATE: 16 Jan 2025

COUNTRY: Azerbaijan

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Equipment: Pump

PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

INCIDENT DESCRIPTION:

A fire occurred at the diesel pumping station caused by a product release due to high vibration from the pump's mechanical seal at an operating temperature of 250-260°C.

WHAT WENT WRONG?

Mechanical seal failure

Excessive vibration

Operating temperature beyond seal capability

Insufficient detection and control

Operating conditions beyond design limits

Inadequate monitoring and preventive maintenance

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Equipment must be operated within design limits

Mechanical integrity is critical

Vibration monitoring is essential

Preventive maintenance prevents escalation

Early detection and alarm systems save lives and assets

Equipment must be operated within design limits

BARRIERS:

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Ignition Control

Hardware Barrier Failures: Protection Systems - including deluge and fire water systems

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 26 Jun 2025

COUNTRY: Azerbaijan

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Workover/Well services

MODE OF OPERATION: Abandonment

POINT OF RELEASE: Wells, drilling and intervention: Well

PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

INCIDENT DESCRIPTION:

A water-clay-gas mixture blowout was observed to erupt from an abandoned onshore well.

WHAT WENT WRONG?

Corrosion and material degradation.

Unexpected reservoir pressure.

Inadequate monitoring and maintenance.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Well abandonment integrity is critical.

Material degradation over time is inevitable.

Unexpected reservoir behavior can occur.

Monitoring is essential.

Strict plugging and sealing standards.

Post-abandonment monitoring.

Risk Assessment before abandonment.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Asset design and integrity

Unspecified: Unspecified

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 13 Jul 2025

COUNTRY: Kazakhstan

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping material/tubing

PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

Contractor personnel were performing a scheduled calibration of a flow meter in accordance with the preventive maintenance plan.

During the calibration process, the technician-initiated pressure stabilization by draining fluid through the designated drainage line. To enhance the drainage process, the technician also opted to use an additional line and proceeded to remove the plug at the end of that line.

Upon removal of the plug, an uncontrolled release of oil containing H₂S occurred. Work was stopped immediately. Contractor workers H₂S detectors activated, donned respirators and evacuated the area.

WHAT WENT WRONG?

The Maintenance Procedure requirement was not followed as Contractor's training was less than adequate for personnel regarding the instrumentation piping schemes, and instrument connection diagrams.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Maintenance Contractors are required to implement training or an annual knowledge assessment, with periodic participation of Company Maintenance representatives, specifically focused on instrumentation piping and hookup diagrams on Company equipment in accordance with Company Maintenance procedures.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 08 Mar 2025

COUNTRY: Kazakhstan

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Breaking Containment Locations: Breaking containment location

PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

A sour gas release occurred at high pressure gas separator. The release occurred while business partners were removing a pressure safety valve for inspection and recertification. While cracking a flange, two workers were exposed to H₂S. As a result, one worker lost consciousness, the other was exposed and became dizzy but remained conscious. The exposed workers were not wearing required Self-Contained Breathing Apparatus.

WHAT WENT WRONG?

Workers were not wearing required breathing apparatus when the H₂S exposure occurred.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Unknown.

BARRIERS:

Hardware Barrier Failures: Life-Saving Equipment - including evacuation systems

CAUSAL FACTORS:

NONE: Unspecified

DATE: 16 Jun 2025

COUNTRY: Kazakhstan

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline

PROCESS SAFETY FUNDAMENTAL: We recognize change.

INCIDENT DESCRIPTION:

At 4:10 pm an emergency operator called to report that security, while patrolling the area, reported a gas smell along the road, at 100m distance from the Security gate. It was a fuel gas release from a 2" FG line during well preparation for MRM.

WHAT WENT WRONG?

Unspecified.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

NONE: Unspecified

RUSSIA & CENTRAL ASIA OFFSHORE

DATE: 14 Dec 2025

COUNTRY: Azerbaijan

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Pipeline operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Subsea: Subsea pipeline/flowline

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Oil patches were observed from the helicopter. Later it was confirmed that a leak had occurred from the subsea 20" pipeline.

WHAT WENT WRONG?

Corrosion and material degradation.

Inadequate monitoring and maintenance.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Material degradation over time is inevitable.

Monitoring is essential.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 12 Jul 2025

COUNTRY: Azerbaijan

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping joint

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

Gas release occurred from 8" blinded flange on TEG contactor due to the degradation of incorrect designed rubber gasket left installed.

WHAT WENT WRONG?

Gas release occurred due to the degradation of a rubber gasket. The gasket was intended solely for transportation and was not replaced with a spiral wound gasket.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Vendor's procedure did not include removing the temporary gasket and project QA/QC was not systematic.

BARRIERS:

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Detection Systems

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

SOUTH & CENTRAL AMERICA ONSHORE

DATE: 01 Oct 2025

COUNTRY: Argentina

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Routine maintenance

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Valve (body, stem, plugs)

PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

Personnel arrive at the gas pipeline in order to uncover two valves. The excavator begins digging until the pipe is identified, then proceeds to remove the metal flag. As the material is removed, the valve fails, releasing gas pressure.

WHAT WENT WRONG?

At the time of the excavation, there was no signalman on site; only the road equipment operator was working.

The integrity condition of the pipeline was degraded.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Define a matrix of excavation tasks, taking into account the necessary resources and safeguards in cases where the standard excavation cannot be met.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

DATE: 22 Oct 2025

COUNTRY: Argentina

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Routine maintenance

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping material/tubing

PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

During fibre optic installation work, an 8" pipe was broken by a backhoe excavator.

WHAT WENT WRONG?

At the time of the excavation, there was no work permit approved for this task.

The work plan did not include excavation equipment.

The risk analysis did not include excavation equipment.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Qualification of staff and contractors in Safe Work Permit.

Implement manual probing in excavation tasks.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PROCESS (CONDITIONS): Protective Systems: Inadequate security provisions or systems

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 17 Nov 2025

COUNTRY: Argentina

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping joint

PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

Gas leak in manifold (measurement bridge), in the plant during production bypass to different gas transport company. Subsequently, the plant shutdown and blowdown were executed, successfully stopping the leak.

WHAT WENT WRONG?

Personnel from the transport company closed their inlet valve without prior notice (refinery also had its valve closed), causing a total blockage of the gas outlet and a sudden increase in pressure in the facilities.

The dielectric joint did not work properly.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

There are no inter-company procedures, defined roles, or communication processes to coordinate critical manoeuvres and/or deliveries between companies.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Human Barrier Failures: Acceptance of handover or restart of facilities or equipment

Management System Element Barrier Failure: Organization, resources and capability

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: 28 May 2025

COUNTRY: Argentina

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Valve (body, stem, plugs)

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

A gas leak occurred as a result of a crack in a nipple on the 4-inch line associated with the globe valve.

WHAT WENT WRONG?

A gas leak occurred as a result of a rupture in a nipple on the 4-inch line associated with the globe valve. A large buoyant gas cloud was observed. SD3 was activated until depressurization through the leak location was achieved.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Ensure that facility design is adequate for the process conditions.

Properly manage changes to facilities by following the defined Management of Change (MoC) practice.

Develop and formalize procedures for operational activities.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

DATE: 14 Mar 2025

COUNTRY: Argentina

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)

PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

During normal operations, a failure occurred in the vent header drainage system, causing liquid carryover to the flare. No material damage or personal injuries were reported; only soil contamination was recorded in the area adjacent to the flare.

WHAT WENT WRONG?

A systemic failure of operational discipline and technical integrity occurred as a result of a chain of latent failures and the normalization of deviation.

At least four (4) layers of protection were identified as having failed or being degraded, with one additional layer compromised. The event was ultimately contained only by the subsequent response of personnel.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Improve the management of operational personnel competencies. Identify the knowledge needs of plant and control room operators. Develop a gap-closure plan that includes training, practical exercises with assessments, and periodic knowledge refreshers.

Strengthen shift handover processes. Standardize, secure, and ensure the proper transfer of critical information related to process conditions, equipment status, abnormal conditions, work permits, and compromised safety barriers.

Reinforce maintenance, inspection, and equipment reliability management. Identify and prioritize process safety-critical equipment, ensuring proper planning, execution, and follow-up of maintenance and inspection activities.

Evaluate and ensure the effectiveness of existing safety barriers. Conduct periodic effectiveness assessments of preventive and mitigative barriers identified in risk studies (HAZOP, LOPA, or others), verifying their availability, independence, and response capability under high-severity scenarios.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 13 Mar 2025

COUNTRY: Brazil

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow

PROCESS SAFETY FUNDAMENTAL: We walk the line.

INCIDENT DESCRIPTION:

Loss of primary containment of petroleum through the safety vent of a tank, visually identified by the area operator. The event occurred during the release of heat exchanger, due to the alignment valve to tank remaining open.

WHAT WENT WRONG?

The alignment valve to tank remained open after the shutdown of the reprocessing pump, allowing the continuous flow of petroleum into the tank until it overflowed.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Operational modifications, even when limited in scope, must be supported by comprehensive risk assessments, covering all possible scenarios and their interfaces with existing systems. The absence of critical steps in operational procedures, such as valve closure after operational manoeuvres, can result in significant operational failures. Governance over non-routine activities needs to be strengthened, with particular attention to Management of Change (MoC), proper use of Operational Instructions (OI), and adequate preparation of the involved team.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

DATE: 05 Dec 2025

COUNTRY: Brazil

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Relief, Vent and Discharge Systems: Drain

PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

During a routine water drainage operation on a LPG Sphere, the drain valve remained open, allowing LPG to be released through this pipeline.

WHAT WENT WRONG?

- Failure to close the drain valve at the end of the outflow.
- Improper use of the drainage system.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

In summary, the lessons learned from this event reinforce the need to:

- Expand the consideration of human factors in all phases of the operational cycle;
- Improve communication and reporting mechanisms, encouraging the active participation of workers;

- Promote periodic reviews of norms and standards, ensuring they are up-to-date, clear, and aligned with the real operational context;
- Strengthen training, simulations, and behavioural verification practices to reduce uncertainties and variations in the execution of critical activities.

BARRIERS:

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 29 Sep 2025

COUNTRY: Brazil

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Unspecified - other

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Valve (body, stem, plugs)

PROCESS SAFETY FUNDAMENTAL: We respect hazards.

INCIDENT DESCRIPTION:

Explosion and fire during a propane leak, resulting in burns to an employee who was performing routine activities in the laboratory. The source of the flammable gas comes from Process Unit, and is used to perform analyses of 100% evaporated LPG residue.

WHAT WENT WRONG?

The needle valve froze during the process and continued to allow passage even after being closed, formation of an explosive atmosphere and ignition of the cloud.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Avoid using process lines inside the laboratory.
- Do not perform the 100% evaporated residue test inside the laboratory.
- Do not dispose of samples in pressurized cylinders inside the fume hood. Disposal must be in an external area.

BARRIERS:

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Ignition Control

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

SOUTH & CENTRAL AMERICA OFFSHORE

DATE: 21 Sep 2025

COUNTRY: Brazil

FUNCTION: Drilling and Completion Operations

NUMBER OF DEATHS: 0

ACTIVITY: Workover/Well services

MODE OF OPERATION: Normal

POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (not at intended discharge location)

PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

During the annulus damping operation, draining the gas from the annulus to the test plant and replacing it with 9.8 ppg inhibited water-based complementary fluid, methane gas was detected on the deck through two gas sensors – one installed internally in the fan house (ventilation room) detected 100% of the LEL (lower explosive limit) of CH₄ and another externally, in an adjacent room, detected 55% of the LEL of CH₄, these rooms that suck air to the stern port column and pontoon of the unit

WHAT WENT WRONG?

1. PTW procedure does not establish appropriate approvers and co-approvers;
2. Instructions to workers not adequate; The PTW form does not reference the document that should be consulted to assist in identifying the equipment as critical;
3. Unspecified simultaneous operations (SIMOP); The PTW procedure does not establish that open PTWs and ongoing operations are evaluated for simultaneity before releasing new PTWs;
4. Lack of operational procedure for ventilation of gases through the lances;
5. The well's WSOG (Well Specific Operation Guidelines) provided for the existence of gas parameters for changing the DP status, being: "advisory" - any gas alarm; and yellow alarm - "confirmed gas", there being no objective definition of the concept of "confirmed gas".

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Review PTW procedure including having the system/equipment end user leader co-approve for work release and train the workforce after review;
2. Train the workforce on the criticality classification of the equipment or system contained in the Service Order linked to the task;
3. Review procedure of "Work Permit System" including clear guidance for assessment of concurrency and approval flow for work permits that were not discussed in the previous day's planning meeting;
4. Include in the risk analysis of the service company's task of operating the Workover Plant the step of checking the alignments and integrity of the manifold and boom lines connected to the plant;
5. Insert a specific line for gas in the WSOG with objective-numeric criteria for "advisory" and yellow alarm.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Human Barrier Failures: Response to emergencies

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Organizational: Inadequate communication

DATE: 07 Mar 2025

COUNTRY: Brazil

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank

PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

INCIDENT DESCRIPTION:

On March 7th, 2025, the platform experienced a major process safety incident when the hinge pin of the dome on a Cargo Tank starboard catastrophically failed. The rupture of the hinge pin resulted in the complete loss of mechanical restraint of the dome, causing the dome cover to fully open and exposing the entire tank opening to atmosphere. This structural failure led to the uncontrolled release of a flammable gas mixture directly onto the deck. The plant's gas detection system was activated, initiating an emergency shutdown (ESD). Personnel were mustered in the accommodation module; however, a continuous flammable gas cloud was extending laterally across the full width of the platform, spanning from one side to the other. As a result, access to the muster location necessarily required personnel to pass through the flammable gas cloud.

The tanks were depressurized both through the fully open dome, which was already providing an unrestricted release path, and by remotely opening the vent stacks. Despite these depressurization measures, the uncontrolled release continued for nearly one hour, until the tanks were fully vented and pressure equalized with the atmosphere. Fortunately, there were no injuries, fires, explosions, or environmental impacts. Nevertheless, production was halted, and personnel were exposed to significant risk due to the prolonged presence of a flammable atmosphere across the installation.

WHAT WENT WRONG?

The root cause investigation identified that the main threat was progressive corrosion on the dome's hinge pin, caused by an inadequate weld that was smaller than specified in the original design. Weaknesses in the preventive safety barriers allowed this defect to persist: the inspection plan did not require detailed checks on the dome pin, and a maintenance anomaly had been incorrectly reported in the system (as a generic defect note instead of a specific inspection note, and linked to a general location instead of the dome pin). These failures in inspection and anomaly management meant the integrity problem was neither prioritized nor properly addressed, allowing the degradation to continue unnoticed until catastrophic failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

The incident highlighted the need for robust inspection and maintenance protocols focused on critical structural joints of tank covers. Inspections must be properly planned and rigorously executed to verify weld quality, dimensional compliance, and integrity at hinge assemblies, pivot points, and attachment areas. Maintenance reporting processes should require precise classification and direct linkage of each finding to the exact component affected, avoiding generic location references. Information management systems must include all relevant structural elements to prevent misregistration or omission of critical observations.

Design and procurement standards should mandate verification of structural joints and weld integrity at commissioning and throughout the operational lifecycle, incorporating these checks into routine field inspections.

Reporting procedures for structural integrity issues within asset management systems must be clearly defined and consistently applied, with personnel retrained to ensure correct categorization and traceability. Existing maintenance records related to tank cover joints should be reviewed and corrected to ensure accurate component-level identification.

Inspection plans must require comprehensive structural assessments of tank covers, including photographic or equivalent documentation to ensure traceability and accountability. In addition, design standards should incorporate secondary locking or securing devices on tank covers and other safety-critical access points to reduce the risk of accidental release or structural failure.

Summary of Main Recommendations:

- Initiate targeted inspections of tank cover hinge joints to confirm weld quality and dimensional compliance.
- Revise inspection checklists to include structural joint verification during acceptance of new tank units and periodically thereafter.
- Clarify and enforce correct asset management reporting categories and retrain inspection and maintenance personnel.
- Audit existing maintenance reports on tank cover areas and correct inaccurate or generic location assignments.
- Update information systems to allow individual selection of all tank cover components when registering anomalies.
- Enhance inspection procedures to require detailed documentation and visual traceability of findings.
- Implement design upgrades with secondary securing devices for tank covers and other safety-critical access points.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Human Barrier Failures: Surveillance, operator rounds and routine inspection

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: 21 Apr 2025

COUNTRY: Brazil

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Unspecified - other

MODE OF OPERATION: Other production

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping material/tubing

PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

INCIDENT DESCRIPTION:

A severe industrial accident occurred on an offshore production platform during gas import operations. At the time, the platform was in the process of decommissioning, with most production systems already shut down and being readied for dismantling; only the gas import line remained operational to supply the power generator. On the morning of the incident, this last active pipeline ruptured after years of progressive corrosion, which developed specifically on its upper surface - an area exposed to salty water dripping through an unsealed opening between decks directly above. Because the line was installed in an elevated position, the most recent inspections did not fully address the upper side, allowing a localized, severe corrosion spot on the top of the pipe to go undetected. When the rupture finally occurred, a large volume of flammable gas was released and quickly ignited, resulting in a powerful jet fire and vapor cloud explosion. Multiple workers were injured, rescue efforts were hampered by equipment failures and communication problems, and the event resulted in significant material damage and evacuation of the platform.

WHAT WENT WRONG?

Several systemic and technical failures contributed to the accident. Key inspections and planned maintenance activities, including the painting and full-circumference inspection of the pipeline, were not prioritized or effectively carried out, partly due to resource constraints and difficulties with access. Procedures for sealing penetrations between decks, established to prevent transfer of water and other substances, were not thoroughly implemented, enabling persistent water leaks onto the pipeline. The corrosion process was not anticipated nor reassessed, despite past records of similar damage in other areas. In addition, operational management did not ensure that the maximum safe pressure for pipeline operation matched actual technical recommendations. Critical barriers such as the automatic firewater system and fire detection did not act fully as designed, prolonging firefighting and evacuation efforts. Emergency communication was hampered both by structural damage and lack of robust back-up systems.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

The accident highlighted the critical importance of robust asset integrity management, disciplined risk assessment processes, clearly defined operational limits, and reliable emergency and communication systems in complex offshore environments. These elements are particularly vital during transitional phases such as decommissioning, when operational priorities shift and legacy assets may operate under reduced monitoring. The event reinforced the need for structured preventive maintenance, integrated risk-based decision-making, strengthened organizational learning, and resilient emergency preparedness. It also emphasized that procedures, responsibilities, and any deferral or cancellation of maintenance must be transparent, technically justified, and consistently documented across all management levels.

Based on these learnings, the key recommendations are:

- Enforce comprehensive inspection regimes, especially in hard-to-access areas, ensuring systemic asset monitoring independent of individual discretion.
- Strengthen governance for prioritization and timely closure of maintenance actions, avoiding backlog accumulation.
- Implement formal controls for sealing and segregation between platform areas, integrating these into design and modification reviews.
- Align operational risk thresholds, such as pressure limits, with updated technical assessments through structured cross-functional reviews.
- Institutionalize periodic reviews of material selection and asset specifications, embedding fire and chemical resistance criteria into procurement standards.
- Maintain strict oversight of automatic detection and protection systems, including testing, availability monitoring, and performance documentation.
- Establish redundant, independent emergency communication channels and routinely verify their resilience.
- Reinforce policies for fire-resistant PPE, including third-party certification and supplier compliance updates.
- Regularly exercise emergency response plans through scenario-based simulations, including degraded communication and multi-asset events.
- Audit inspection, documentation, and information management systems to ensure accuracy, clear accountability, and organizational learning.
- Develop scenario-driven protocols for escape route contingency management with risk-based criteria for temporary impairments.
- Strengthen health and rescue response capabilities, integrating staffing strategies and new technologies for remote sites.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Hardware Barrier Failures: Protection Systems - including deluge and fire water systems

Human Barrier Failures: Surveillance, operator rounds and routine inspection

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Overexertion or improper position/posture for task

PEOPLE (ACTS): Use of Protective Methods: Personal Protective Equipment not used or used improperly

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Work Place Hazards: Congestion, clutter or restricted motion

DATE: 22 Apr 2025

COUNTRY: Brazil

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Pipeline operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Subsea: Subsea pipeline/flowline

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

During routine gas injection operations on an offshore production unit, a sudden drop in pressure and an increase in flow were detected in one of the flexible gas injection lines. Upon investigation, it was discovered that the line had ruptured at a deep-water connector, leading to an uncontrolled release of flammable gas mixture. The event did not result in injuries or environmental impact but caused significant material loss. Following the detection, the gas injection was manually stopped to contain the release, and remote inspections confirmed the subsea failure.

WHAT WENT WRONG?

The event originated from a failure in the flexible pipeline at a connector that had been assembled without specific mounting controls. The failure mechanism involved stress corrosion cracking induced by CO₂, made possible or accelerated due to several compounding factors: inadequate risk differentiation for connectors assembled without precise tools, use of a less conservative methodology for determining component lifespan, and high mechanical stress associated with field conditions. Critically, a key automatic safety shutdown system had been intentionally disabled for operational reasons and was not reactivated—a gap in procedural compliance and communication, delaying the response and increasing the volume released.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

The main lessons centre on the need for robust integrity management systems, effective barrier monitoring, clear operational interfaces between different teams, and consistent risk evaluation practices. The event underscored that comprehensive controls, adherence to defined procedures, and thorough risk communication are essential for preventing escalation, even in the presence of complex technical threats.

Key recommendations and management learnings:

- Adopt stricter controls regarding the differentiated risk of assembling connectors without precision mounting tools, integrating this consideration into the overall integrity management system.
- Ensure that risk acceptance criteria for lifetime management are appropriately conservative, especially when dealing with known threat mechanisms—periodically revalidate methodologies and revisit underlying assumptions.
- Implement and enforce clear procedures for disabling and reactivating safety barriers, particularly those associated with interdependent operational areas; ensure all inactivations have documented risk assessment and multidisciplinary review.
- Reinforce monitoring routines for all inhibited safety systems, including automated notifications and prompt follow-up to minimize periods of exposure.
- Strengthen operational interface and communication protocols between onshore support, oversight, and field personnel to guarantee mutual awareness and immediate response capability when critical safety systems are inhibited or restored.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 24 Jul 2025

COUNTRY: Brazil

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Relief, Vent and Discharge Systems: Discharge to sea

PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

INCIDENT DESCRIPTION:

During a fluid recirculation operation between storage tanks, an unintended release of crude oil occurred into the sea. The operation aimed to measure hydrogen sulfide levels present in a slop tank, routing the fluid through an analyzer. It was later discovered that oil from the oil phase from the tank, had passed through the discharge system due to the failure in a valve, resulting in a significant environmental impact.

WHAT WENT WRONG?

The root cause of the event was a mechanical failure: the centering bushing inside the actuator shaft suffered damage, physically preventing the valve from fully closing. Additionally, operational procedures were incomplete, as they did not ensure proper interface level measurement prior to recirculation. The valve acted as the only barrier in the discharge system, and there was a history of reliability issues with similar components. The overall system lacked redundancy, and certain risk scenarios had not been fully considered or mapped in risk analysis routines.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Generically, this event underscored the need to strengthen management controls that ensure reliability of safety barriers and completeness of operational procedures, as well as to promote systematic periodic review of maintenance plans and risk management routines. It also reinforced the importance of learning from past failures to optimize organizational preventive measures.

Systemic management lessons & recommendations:

- Adopt management controls to ensure systematic review of the adequacy and reliability of shutdown devices, validating that replacements or upgrades address all known failure modes.
- Regularly update maintenance plans to require direct inspection and functional verification of components beyond indicator readings—in particular, confirming actuator movements visually during testing.
- Revise operational procedures to guarantee checks and measurements of interface levels in storage tanks prior to recirculation or discharge operations, preventing unintentional product releases.
- Incorporate requirement of additional barriers or redundancies in discharge systems, so that a single-point failure does not result in environmental incidents.
- Establish periodic technical reviews of risk scenarios, ensuring changes in operating routines are mapped into risk analyses and risk mitigation measures.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

DATE: 24 Aug 2025

COUNTRY: Brazil

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Subsea: Subsea equipment

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

A significant release of flammable gas occurred in a subsea production system when the small-diameter pipe responsible for chemical injection instrumentation failed completely. The incident went initially unnoticed by automatic safety systems. The escape of gas was eventually discovered during scheduled activities using remotely operated equipment for another purpose. The event resulted in a discharge of a large volume of gas into the sea; no injuries or direct material impact were reported.

WHAT WENT WRONG?

The root cause of the incident was mechanical fatigue of the chemical injection tubing, caused by vibration induced by gas flow over time. At the design stage, there were no requirements or practices for assessing the risk of vibration on such instrumentation lines attached to the pressure containment body. As a result, the primary containment barrier for the chemical injection line failed due to degradation from vibration in a so-called “dead leg” configuration. Contributing to the incident, the automated safety and shutdown systems did not detect the event: the variation in operational pressure was too small to trigger shutoff actions. The shutdown was performed manually after detection.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

This incident demonstrates the importance of systematically managing vibration risks in all pressurized instrumented lines, including those not traditionally subject to such analysis. It highlights the necessity to embed vibration assessment requirements in the technical specifications and project phases for new and existing installations. Strengthening the design and verification process helps to ensure the integrity of barriers and timely detection of degradation mechanisms that automated process systems may not catch.

Key recommendations derived from the case include:

- Review and update technical requirements for new projects so that analysis for vibration effects, including acoustic and flow-induced vibration, is explicitly required for all instrumented lines attached to the main process body.
- For systems currently under construction, ensure that vibration studies are extended to cover these instrument tubing lines before commissioning.
- For systems already in operation, develop engineering improvements and plan the implementation to ensure that all elements subject to comparable service conditions are covered for integrity over their life cycle.
- Calibrate and implement analytical tools to be capable of estimating the time to failure due to vibration fatigue in small-bore instrument lines.
- Based on these studies, create specific action plans for each affected facility, including possible reduction of operating parameters, scheduled removals, or engineering retrofits needed to guarantee long-term reliability.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 18 Sep 2025

COUNTRY: Brazil

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Routine maintenance

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Piping joint

PROCESS SAFETY FUNDAMENTAL: We apply procedures.

INCIDENT DESCRIPTION:

During a planned maintenance activity to replace fasteners on a gas injection pipeline, a high-pressure release of a flammable gas mixture occurred. The release happened as the maintenance crew attempted to perform a sequence of 'hot bolting,' a procedure involving the removal and replacement of bolts on a flange while the pipeline remained pressurized. Due to failures in procedural adherence and execution, the flange opened unexpectedly, resulting in a jet of cold, pressurized gas that struck workers, causing eye injuries and requiring urgent medical evacuation.

WHAT WENT WRONG?

The immediate cause of the incident was the inability to maintain the minimum torque required for flange tightness. Critical errors included the use of incorrect tools, performing the task in an improper sequence, and a lack of adequate qualification among those involved. Procedures lacked clear guidance on risk assessment and practical execution, with insufficient focus on detailed planning and inspection prior to the task. Administrative safeguards failed, particularly in terms of controlling the tools used and ensuring workers possessed the specific technical competence demanded by both the process and normative standards. While layered protections such as emergency shutdown and gas detection responded effectively to limit escalation, preventive measures did not.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

This event underscored that high-criticality tasks like hot bolting demand strict compliance with technical standards, practical procedures that are clearly defined and executable, and that relying solely on administrative controls is inadequate. The organization must ensure that risk assessments and planning encompass real working conditions and that exposure to hazardous situations is controlled by robust physical, engineering, and organizational safeguards rather than individual choices.

Main recommendations derived from the event were:

Revise technical standards for hot bolting to ensure that all engineering risk assessment items are addressed during planning, documented and implemented in execution.

Update procedures so they fully comply with all normative requirements, and reinforce the need for strict adherence in execution—as deviations may lead to uncontrolled loss of containment.

Adapt the permit to work system to explicitly prohibit unsuitable tools (such as hammers) for hot bolting, incorporating regulatory restrictions.

Improve hot bolting services by specifying professional qualifications required, and revise contract requirements to ensure only appropriately trained personnel are assigned to such critical tasks.

Increase the categorization of permits for hot bolting, demanding more rigorous analysis during planning.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 22 Oct 2025

COUNTRY: Brazil

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Pipeline operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Subsea: Subsea pipeline/flowline

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

At approximately 05:45, a rupture occurred in a gas lift riser pipeline. Gas detectors on the unit identified the presence of a gas cloud, which triggered an emergency shutdown of the facility, resulting in the immediate halt of production operations. Personnel were promptly relocated to designated muster points as a precautionary measure.

Firefighting and oil recovery vessels were deployed to the site for monitoring and response. An emergency response structure was mobilized at the operations center. A subsea construction vessel equipped with a remotely operated vehicle (ROV) was assigned to inspect the risers on the offshore platform to identify the source of the leak.

Continuous monitoring of gas sensors confirmed that the gas detection had ceased. The rupture in the gas lift riser is currently under investigation. Further laboratory analysis of the damaged riser is required to determine the root cause, and the event remains under investigation pending these results.

WHAT WENT WRONG?

Under investigation.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Under investigation.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

NONE: Unspecified

DATE: 08 Nov 2025

COUNTRY: Brazil

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Breaking Containment Locations: Breaking containment location

PROCESS SAFETY FUNDAMENTAL: We recognize change.

INCIDENT DESCRIPTION:

A chemical operator was performing a transfer and dosing operation of a caustic product into a cooling water system in an industrial plant. During the disconnection of a hose used for product injection (in an improvised location), residual pressure caused the chemical to splash onto the operator's face and over his protective coverall. In reaction, the operator jumped from a portable ladder (about 1.5 meters high), resulting in a twisted ankle in addition to the chemical burn. The worker received immediate first aid and required medical attention, leading to time off work.

WHAT WENT WRONG?

What went wrong was that the task was performed using an improvised procedure, without any specific instructions or operational manuals covering the scenario. Deficiencies were evident in the management of operational changes, as there was no dedicated plan or risk analysis in place for the modified chemical injection point. The operator was using regular personal protective equipment, but did not have the specific goggles recommended for handling this chemical. Furthermore, communication regarding system changes and status was insufficient, resulting in misunderstandings about equipment availability and the associated risks. Maintenance backlogs forced the adoption of alternative and less safe methods, thus exposing workers to additional hazards. Additionally, there was a lack of structured practical training for new or less experienced operators, and critical tasks were carried out alone, without verification or support from a second trained person.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

This event underscored the need for strong management of operational changes, effective communication, risk analysis, and continuous improvement of procedures and training to prevent recurrence of such incidents.

Key recommendations:

- Formalization of operational procedures for selected critical activities, including structured risk analysis.
- Reinforcement of internal communication to ensure any system changes are fully understood by all involved.
- Implementation of structured practical training for new operators, emphasizing human factors and safe execution.
- Creation of formal mechanisms that enable employees to report uncertainties and safely interrupt activities in case of doubt.
- Promotion of leadership workshops to reinforce the requirement for opening and managing changes for any operational deviations.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures - PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PEOPLE (ACTS): Use of Protective Methods: Personal Protective Equipment not used or used improperly

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

DATE: 18 Dec 2025

COUNTRY: Brazil

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Valve (body, stem, plugs)

PROCESS SAFETY FUNDAMENTAL: Not applicable.

INCIDENT DESCRIPTION:

During the maintenance activity to remove instrumentation from a gas export pipeline, a manual valve stem detached, leading to the release of flammable gas. The gas leak was detected by multiple gas sensors, prompting a manual emergency shutdown of the facility, which included depressurization procedures and safe evacuation of personnel. No injuries or environmental damage occurred, and the incident was managed according to emergency response protocols.

WHAT WENT WRONG?

The release resulted from multiple management and technical failures. Inspection and maintenance procedures lacked clear instructions for properly evaluating small-diameter valves, which made the detection of advanced corrosion difficult. Additionally, there was inadequate monitoring of painting quality and corrosion protection, particularly in components made of dissimilar materials. This deficiency led to galvanic corrosion and rapid degradation of the valve body. Furthermore, material specification standards did not prevent the use of dissimilar materials in new installations, contributing to additional corrosion risks. Finally, the facility project lacked a double isolation mechanism, allowing gas to pass through manual valves and consequently increasing the volume released.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

This incident highlights the importance of thorough management systems for integrity and inspection of pipeline accessories, especially small-diameter valves and those involving dissimilar materials. It reinforced the need for robust procedures, quality control, and adherence to engineering best practices across similar operations.

Recommendations:

- Revise technical specifications and procedures to ensure they address risks associated with dissimilar materials.
- Conduct technical workshops for inspection teams on failure mechanisms in small-diameter valves (mainly dissimilar materials).
- Establish integrity evaluations for valves in similar operational conditions and replace those showing signs of galvanic corrosion.
- Prohibit the use of dissimilar materials in valves for new projects.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Human Barrier Failures: Surveillance, operator rounds and routine inspection

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

NONE: Unspecified

DATE: 14 Sep 2025

COUNTRY: Guyana

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Start-up

POINT OF RELEASE: Breaking Containment Locations: Sample system

PROCESS SAFETY FUNDAMENTAL: We stop if the unexpected occurs.

INCIDENT DESCRIPTION:

Night shift production supervisor was assisting projects instrumentation persons to put probe of Test separator outlet base sediment and water analyser in place as part of the equipment commissioning. During the operation, as reported in his statement all steps as per manual was followed. Unfortunately he encountered a few issues, the probe was very hard to handling/move. Finally, he decided to come down the stepladder and the same time the probe gland and trail dislodged from its original position resulting in crude oil leak on deck and splashed on the production supervisor's body.

WHAT WENT WRONG?

During night shift commissioning activities for the base sediment and water analyser on the test separator outlet, the Production Supervisor assisted instrumentation personnel in positioning the probe. Despite following the manual, the probe proved difficult to manoeuvre. While descending a stepladder, the probe gland and trail unexpectedly dislodged, resulting in a crude oil leak on the deck and splashing onto the supervisor. Immediate safety protocols were activated, including gas detection triggering the PAGA alarm, manual ESD activation, crew muster, and plant blowdown. Once gas levels subsided, the Emergency Response Team (ERT) secured the area, and clean up operations were conducted.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

DATE: 22 Jan 2025

COUNTRY: Trinidad and Tobago

FUNCTION: Production

NUMBER OF DEATHS: 0

ACTIVITY: Production operations

MODE OF OPERATION: Normal

POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea): Choke

PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

INCIDENT DESCRIPTION:

The release occurred on the well choke valve due to sand erosion. The combination of high water and sand production introduced complex fluid dynamics within the choke leading to erosion of the plug, preferential flow across the trim and erosion of the choke's outlet body.

WHAT WENT WRONG?

High sand caused erosion on a well choke valve creating a hole in the line.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Establish risk-management processes for high-risk sandy wells, including formal assessments, documentation of enduring risks, and mandatory consultation with sand SMEs.

Assess sand detection monitoring and algorithms can enhance process control by providing alarms and enabling automated shut-in of wells or platforms.

Review the Fire & Gas logic to ensure shutdown functions meet company requirements.

Implement an improved alarm-response plan with updated procedures for gas-detector alarms, ASD overrides, alarm shelving, and KPI reviews aligned with process-safety improvement recommendations.

Update Hub and NUI Emergency Response Plans to ensure accurate, ARM-aligned responses to gas-detector alarms, with clear distinctions between manned and unmanned scenarios.

BARRIERS:

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Detection Systems

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

FATAL INCIDENTS CLASSIFIED AS TIER 1 PSE

In this Section, fatal incidents reported in the safety performance dataset that were also reported as Tier 1 PSE in the process safety data set are presented.

AFRICA OFFSHORE

DATE: 20 May 2025

COUNTRY: Angola

NUMBER OF DEATHS: 4

NUMBER OF PERMANENT IMPAIRMENT INJURIES: 4

FUNCTION: Production

CAUSE: Explosion, fire or burns

ACTIVITY: Maintenance, inspection, testing

PRIMARY LIFE-SAVING RULE: Hot work

SECONARY LIFE-SAVING RULE: Energy isolation

FATALITY

Function: Construction, Employer: Contractor, Occupation: Unknown, Body part: Head (incl. mouth, etc.), Nature of injury: Burn, Time in service: -

FATALITY

Function: Construction, Employer: Contractor, Occupation: Unknown, Body part: Internal organs, Nature of injury: Burn, Time in service: -

FATALITY

Function: Construction, Employer: Contractor, Occupation: Unknown, Body part: Internal organs, Nature of injury: Burn, Time in service: -

FATALITY

Function: Construction, Employer: Contractor, Occupation: Unknown, Body part: Internal organs, Nature of injury: Burn, Time in service: -

PERMANENT IMPAIRMENT

PI Category: Unspecified, PI Subcategory: Unspecified, Body part: Head (incl. mouth, etc.), Nature of injury: Burn, Time in service: -, Function: Construction, Employer: Contractor, Occupation: Unknown, Body part: Head (incl. mouth, etc.), Nature of injury: Burn, Time in service: -

PERMANENT IMPAIRMENT

PI Category: Unspecified, PI Subcategory: Unspecified, Body part: Head (incl. mouth, etc.), Nature of injury: Burn, Time in service: -, Function: Construction, Employer: Contractor, Occupation: Unknown, Body part: Head (incl. mouth, etc.), Nature of injury: Burn, Time in service: -

PERMANENT IMPAIRMENT

PI Category: Unspecified, PI Subcategory: Unspecified, Body part: Head (incl. mouth, etc.), Nature of injury: Burn, Time in service: -, Function: Construction, Employer: Contractor, Occupation: Unknown, Body part: Head (incl. mouth, etc.), Nature of injury: Burn, Time in service: -

PERMANENT IMPAIRMENT

PI Category: Unspecified, PI Subcategory: Unspecified, Body part: Head (incl. mouth, etc.), Nature of injury: Burn, Time in service: -, Function: Construction, Employer: Contractor, Occupation: Unknown, Body part: Head (incl. mouth, etc.), Nature of injury: Burn, Time in service: -

NARRATIVE:

Crews were performing flange management on the water 16" and 18' lines and installation of the blind flange on the 24' overboard line for hydrotest. During the process, crew encountered a clash with an abandoned open 18' line and started hot cutting the 18' line to make room to install a blind flange. Fire broke out at worksite, resulting in 17 people injured (including first aid cases) and 1 person missing.

WHAT WENT WRONG:

Fire or burn.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Pending Investigation.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

ASIA/AUSTRALASIA ONSHORE

DATE: 07 Jan 2025

COUNTRY: India

NUMBER OF DEATHS: 1

FUNCTION: Unspecified

CAUSE: Explosion, fire or burns

ACTIVITY: Maintenance, inspection, testing

PRIMARY LIFE-SAVING RULE: Hot work

SECONDARY LIFE-SAVING RULE: Line of fire

FATALITY

Function: Production, Employer: Company, Occupation: Engineer, Scientist, Technician, Body part: Respiratory system, Nature of injury: Burn, Time in service: 10 + years

NARRATIVE:

Fatal fire following premature restart during PSV maintenance.

WHAT WENT WRONG:

A premature restart of the system without verifying reinstatement of the PSV resulted in uncontrolled hydrocarbon release, gas accumulation, ignition from a modified adjacent burner opening, and a fatal fire incident.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Critical safety devices must never be removed or bypassed without ensuring positive mechanical isolation, structured restart control, and thorough hazard assessment, particularly in hydrocarbon environments with adjacent operating units.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 31 Aug 2025

COUNTRY: Indonesia

NUMBER OF DEATHS: 1

FUNCTION: Drilling

CAUSE: Struck by (not dropped object)

ACTIVITY: Drilling, workover, well operations

PRIMARY LIFE-SAVING RULE: Energy isolation

SECONDARY LIFE-SAVING RULE: Line of fire

FATALITY

Function: Drilling, Employer: Contractor, Occupation: Drilling/well servicing operator, Body part: Head (incl. mouth, etc.), Nature of injury: Organ function / loss or damage, Time in service: > 5 years < 10 years

NARRATIVE:

Worker died due to (allegedly) being hit by a surface test line while carrying out well surveillance work.

WHAT WENT WRONG:

Safety restraint/whip check in substandard conditions.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Ensure pressure test complies with procedures.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

ASIA/AUSTRALASIA OFFSHORE

DATE: 22 Oct 2025

COUNTRY: Malaysia

NUMBER OF DEATHS: 1

FUNCTION: Production

CAUSE: Struck by (not dropped object)

ACTIVITY: Production operations

PRIMARY LIFE-SAVING RULE: Bypassing safety controls

SECONDARY LIFE-SAVING RULE: Line of fire

FATALITY

Function: Production, Employer: Contractor, Occupation: Maintenance, craftsman, Body part: Head (incl. mouth, etc.), Nature of injury: Concussion, Time in service: > 1 year < 5 years

NARRATIVE:

On 22 October 2025 at approximately 1615 hours, smoke alarm detectors were triggered in the engine room. Personnel on Board (POB) mustering and engine room reported one missing person. Emergency Response Team (ERT) deployed to the location and found one engine room operator unconscious, adjacent to steam plant. Additionally, there was sign of steam release found at site. Despite prompt medical attention by the onboard medical team, the individual was subsequently pronounced deceased. The affected equipment and piping were isolated.

WHAT WENT WRONG:

This incident was due to the following factors:

- a. Condensation-induced water hammer
- b. The failure of the end plate weldment
- c. The end plate was ejected and hit the DP on the right side of his head

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Gaps in system operation, monitoring, and competency management. This serves as an important reminder of the need for strong technical control and workforce awareness when operating steam systems.

This incident highlights the importance of ensuring all crew, especially new and existing personnel, are properly trained and aware of steam system risks such as steam hammering, and are competent in following approved operating procedures. Critical equipment including control valves, steam traps, and piping must be maintained in proper working condition and operated in automatic mode where required to prevent condensate build-up and system damage. Adequate temperature and pressure monitoring must be in place and closely observed so that early intervention can be taken during abnormal conditions. Operating procedures must be regularly reviewed and updated to include steam hammering mitigation measures, and the Management of Change (MoC) process must be strengthened to ensure that any system modifications are properly assessed, documented, and reflected in as-built drawings.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PEOPLE (ACTS): Use of Protective Methods: Disabled or removed guards, warning systems or safety devices

NORTH AMERICA ONSHORE

DATE: 10 Dec 2025

COUNTRY: Mexico

NUMBER OF DEATHS: 1

FUNCTION: Unspecified

CAUSE: Struck by (not dropped object)

ACTIVITY: Production operations

PRIMARY LIFE-SAVING RULE: Line of fire

FATALITY

Function: Unspecified, Employer: Contractor, Occupation: Maintenance, craftsman, Body part: Leg/knee/hip, Nature of injury: Cut, Puncture, Scrape, Time in service: -

NARRATIVE:

The contractor worker, 50 years old, while performing activities on an 18" floating gate valve located at the inlet line to separator at the Gas Transportation Processing Centre, was struck by a metal element in the right front hip. He was transported to the clinic located outside the facility, where the on-duty physician declared him deceased at 10:30 h.

WHAT WENT WRONG:

The incident was originated from the failure of an O-ring. This failure caused the violent ejection of the worm screw, resulting in a fatal injury to a worker. This finding highlights a systemic deficiency in the certification and quality processes for replacement components used in Floating Gate Valves (FGVs).

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Contractors must provide evidence of maintenance, inspection, testing, and certification for all machinery and equipment they will use, ensuring that tests comply with applicable regulations. Through a temporary management of change process, it must be confirmed that contractor equipment does not introduce additional risks. The valve design must comply with API-6D, and material certifications must meet NACE MR-0175 requirements for sour service. The procedure must also be updated to include the correct personnel positioning during valve opening and the actions to take in case of abnormal conditions.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 18 Sep 2025
COUNTRY: USA
NUMBER OF DEATHS: 1
FUNCTION: Construction
CAUSE: Struck by (not dropped object)
ACTIVITY: Construction, commissioning, decommissioning
PRIMARY LIFE-SAVING RULE: Energy isolation
SECONARY LIFE-SAVING RULE: Line of fire

FATALITY

Function: Construction, Employer: Contractor, Occupation: Unknown, Body part: Head (incl. mouth, etc.), Nature of injury: Concussion, Time in service: -

NARRATIVE:

Around 10 minutes into clearing/removal activities a high energy release occurred ejecting the contents of the piping assembly including the pig, fatally striking a contract worker in the head.

WHAT WENT WRONG:

High energy release.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Pending Investigation.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

FATAL TIER 1 PSE NOT INCLUDED IN THE SPECIFIC PSE DATA COLLECTION

Participation in the IOGP annual collection of safety and process safety performance data is voluntary, and not all Member Companies participate in each specific data set. In 2025 all fatal PSE were included in both data sets.

FATAL INCIDENTS RELATED TO PROCESS SAFETY

In this Section, fatal incidents reported in the safety performance dataset that were related to process safety but were not classified as PSE are presented.

MIDDLE EAST ONSHORE

DATE: 08 Apr 2025

COUNTRY: Kuwait

NUMBER OF DEATHS: 1

FUNCTION: Production

CAUSE: Explosion, fire or burns

ACTIVITY: Production operations

PRIMARY LIFE-SAVING RULE: Other issue – no applicable rule

FATALITY

Function: Production, Employer: Contractor, Occupation: Foreman, Supervisor, Body part: Internal organs, Nature of injury: Organ function / loss or damage, Time in service: -

NARRATIVE:

For the fuel gas pipeline was under preparation for cleaning pigging activity, contractor employees went to the location for cleaning pigging activity. During the process, an explosion (without fire) occurred, sand and carbon were scattered in the area. This led to the collapse of the valve pit along with its structure, resulting in one contractor employee fatality, one contractor employee injury (left leg ankle fracture) and damage to the property. The injured employees were shifted to company clinic and then to hospital. One contractor employee was declared dead by the hospital and another contractor employee was injured.

WHAT WENT WRONG:

1. An explosion of an underground pipeline section because of loss of containment.
2. The loss of containment resulted in gas accumulation at congested and confined underground section of the pipeline next to the concrete pit that was ignited by an exothermic reaction between pyrophoric black powder and air.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Upgrade old manifolds pigging facilities to ensure safety while carrying out pigging operations.
2. Ensure Safe Deferral process is implemented for the deferment of pipeline inspections.
3. Track Inspection & Corrosion recommendations and prioritized based on risk.
4. Ban the use of Generic Job Safety Analysis and consider it as a violation.
5. The use of e-Work Permit System shall be enforced at all times, unless authorized by HSE Team.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

HIGH POTENTIAL EVENTS CLASSIFIED AS TIER 1 PSE

In this Section, high potential events reported in the safety performance dataset that were also reported as Tier 1 PSE in the process safety data set are presented.

ASIA/AUSTRALASIA ONSHORE

DATE: 17 Jul 2025

COUNTRY: Thailand

FUNCTION: Production

CAUSE: Explosion, fire or burns

ACTIVITY: Production operations

PRIMARY LIFE-SAVING RULE: Hot work

NARRATIVE:

Fire incident occurred at the lube oil breather valve on the crankcase of Gas Lift Compressor.

WHAT WENT WRONG:

- The working procedure did not follow the manufacturer's instructions and lacked clear steps for torque rechecking and proper packing ring installation while the compressor was running.
- The risk assessment failed to consider a potential high-pressure gas leak into the crankcase.
- There was no fire detection or automatic blowdown system, and gas detection was insufficient.
- Gaps in insulation allowed gas to contact hot turbocharger surfaces.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Procedures must strictly follow manufacturer requirements and include clear, detailed steps with proper staff training.
- Risk assessments should address all credible leak scenarios.
- Adequate gas detection, fire detection, and automatic shutdown/blowdown systems must be installed.
- Equipment insulation and monitoring systems should be regularly checked to prevent gas contact with hot surfaces.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

EUROPE ONSHORE

DATE: 11 Oct 2025

COUNTRY: UK

FUNCTION: Production

CAUSE: Explosion, fire or burns

ACTIVITY: Production operations

PRIMARY LIFE-SAVING RULE: Bypassing safety controls

SECONARY LIFE-SAVING RULE: Work authorization

NARRATIVE:

Loss of Primary Containment – Methanol Release.

WHAT WENT WRONG:

Inadequate mental of physical ability/ergonomic.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

None issued.

CAUSAL FACTORS:

No causal factors allocated.

MIDDLE EAST ONSHORE

DATE: 26 Aug 2025

COUNTRY: Kurdistan Region Of Iraq

FUNCTION: Production

CAUSE: Exposure noise, chemical, biological, vibration, extreme temperature

ACTIVITY: Production operations

PRIMARY LIFE-SAVING RULE: Other issue – no applicable rule

NARRATIVE:

An incident occurred at NGL Loading Bay involving the overfilling of a truck and subsequent spill of NGL. The event resulted from the failure of the High-High level switch, which did not initiate the automatic system shutdown as designed. The abnormal condition was identified through CCTV monitoring, which revealed the product overflow. Immediate intervention was undertaken by activating the emergency shutdown push button, thereby halting loading operations.

WHAT WENT WRONG:

1. Failure of Primary Level Control Barrier - high-high level switch failed to operate and did not initiate the automatic closure of the associated loading valves.
2. Design Flaws in Timer-Based Overfill Protection - The tertiary overfill protection relied on a preset loading timer based on a fixed batch concept, which did not account for varying tanker sizes or flow rates.
3. Degraded Condition of Critical Equipment - The level switch assembly and cabling were found to be frequently damaged due to poor ergonomic design of the vapor extraction and switch mounting arrangements, and mechanical interference during tanker installation and removal.

4. Redundant Barrier Not Maintained in Practice - A local high-level switch, previously identified as a safety barrier in earlier designs and referenced in the Control and Operating Philosophy, was retained physically but removed as a functional barrier during the re-HAZOP.

5. Procedural and Human Factors Weaknesses - Although pre-loading checks are required, they did not adequately account for the degraded condition of key safeguards.

The dependency on a single barrier and the lack of operator awareness about available fallback systems further compounded the risk.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Remove the logic for the end of sequence timer protection and remove its reference from the Control and Philosophy procedure.
2. Carry out an ergonomic design assessment of the Hatch Cover assembly
3. Incorporate the local high-level Alarm as the secondary barrier.
4. Conduct a comprehensive Safety Critical Task Analysis's (SCTA) of the NGL road tanker loading operation.
5. Develop detailed Standard Operating Procedures (SOP) for the NGL/LPG and tanker loading process.
6. Deliver a training program to all affected stakeholders.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

NORTH AMERICA ONSHORE

DATE: 20 May 2025

COUNTRY: Canada

FUNCTION: Production

CAUSE: Explosion, fire or burns

ACTIVITY: Maintenance, inspection, testing

PRIMARY LIFE-SAVING RULE: Energy isolation

SECONDARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

On May 20, 2025, a worker was engaged in repositioning a spec blind from the closed to the open position on the flare line, in preparation to de-isolate the unit for commissioning activities. During this activity, the worker was exposed to a gas/energy release from the down steam side of the isolated gate valve at the open flange pair. Worker required rescue from heights and was provided onsite first aid; worker was transported to a local hospital for further medical assessment where it was determined worker sustained bruised ribs and a concussion. Worker was released from hospital the same day.

WHAT WENT WRONG:

The 10" gate valve used as a single isolation was not fully closed because the chain operator seized due to physical interference, giving a false indication that the valve was seated. Pressure verification showed 0 kPag, but this was ambiguous because the flare system was under slight vacuum, meaning zero pressure did not confirm positive isolation. Oxygen entered the flare piping system through the partially open valve while the flare remained lit,

creating conditions for ignition. When work was performed on the spectacle blind, this led to a deflagration and an uncontrolled release of energy that injured the worker.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Zero pressure does not equal positive isolation - Pressure indication alone is an insufficient barrier for verifying isolation in flare systems, especially where vacuum conditions can exist.
2. Single isolation is not a robust barrier for breaking containment - Single isolation provides little tolerance for hidden failures and should not be relied upon where severe energy release is credible.
3. Equipment condition and work environment directly shape human decision-making - People act on the cues the system gives them; degraded equipment and obstructed layouts can actively mislead competent workers.
4. Permits and risk classification must reflect real hazard potential - Administrative controls lose credibility when they do not reflect the true severity of the hazard.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 28 Feb 2025

COUNTRY: Canada

FUNCTION: Production

CAUSE: Exposure noise, chemical, biological, vibration, extreme temperature

ACTIVITY: Transport - Land

PRIMARY LIFE-SAVING RULE: Line of fire

SECONDARY LIFE-SAVING RULE: Energy isolation

NARRATIVE:

During a fluid-hauling task, water transfer to a tank truck was completed and the hose was disconnected. A valve was found not fully closed, and hot fluid sprayed during subsequent valve manipulation. The job was stopped, emergency medical response was initiated, and the well/site was shut down to support response and control the release.

WHAT WENT WRONG:

Residual energy remained in the system at the time of disconnection/valve manipulation, resulting in a hot-fluid spray. The investigation identified gaps in procedural requirements for verifying zero energy and specifying chemical PPE for hot-fluid tasks, and noted that the system configuration did not readily allow field verification of “zero energy.”

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Update procedures and minimum PPE requirements for hot-fluid tasks (including defined thresholds) and incorporate explicit “verify zero energy” expectations. Where verification is not practicable, implement engineering/administrative improvements (e.g., bleed/verification points or safer valve/coupler configurations) so that residual energy can be confirmed and controlled before disconnecting equipment.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PEOPLE (ACTS): Use of Protective Methods: Personal Protective Equipment not used or used improperly

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 26 May 2025

COUNTRY: USA

FUNCTION: Drilling

CAUSE: Pressure release

ACTIVITY: Drilling, workover, well operations

PRIMARY LIFE-SAVING RULE: Energy isolation

SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

Crew was killing the well with Axis Reverse Unit when the Axis Pump Hand appears to have engaged valves in the incorrect order allowing pressure to exit well bore through a disconnected 2" hose at the rig floor. Pressure estimated to be 600 psi. Loose end of hose whipped and struck one individual on the top of their hard hat.

WHAT WENT WRONG:

Axis pump hand opened a valve and allowed wellbore pressure to energize the hose resulting in it whipping and striking the IP.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Improper sequence of steps (valve alignment) which resulted in an unintended release of pressure to the kelly hose striking the Floorhand.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 01 Aug 2025

COUNTRY: USA

FUNCTION: Production

CAUSE: Explosion, fire or burns

ACTIVITY: Construction, commissioning, decommissioning

PRIMARY LIFE-SAVING RULE: Work authorization

SECONARY LIFE-SAVING RULE: Confined space

NARRATIVE:

A flash fire occurred during the demolition of internal components in the first-stage knockout drum (module C2) at the facility.

WHAT WENT WRONG:

Work performed under confined space entry and hot work permits occurred more than 2 hours after permits were initially issued and was not revalidated prior to beginning work.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

All permits must be revalidated if work is not started within a maximum of 2 hours of issue or if there is a break in the work of 2 hours or more. Makes necessary tests for flammable and/or other hazardous conditions immediately prior to the start of hot work or when work is suspended for more than 2 hours.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

RUSSIA & CENTRAL ASIA ONSHORE

DATE: 13 Jul 2025

COUNTRY: Kazakhstan

FUNCTION: Production

CAUSE: Exposure noise, chemical, biological, vibration, extreme temperature

ACTIVITY: Maintenance, inspection, testing

PRIMARY LIFE-SAVING RULE: Energy isolation

SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

Contractor personnel were performing a scheduled calibration of a flow meter in accordance with the preventive maintenance plan.

During the calibration process, the technician initiated pressure stabilization by draining fluid through the designated drainage line. To enhance the drainage process, the technician also opted to use an additional line and proceeded to remove the plug at the end of that line.

Upon removal of the plug, an uncontrolled release of oil containing H₂S occurred. Work was stopped immediately. Workers H₂S detectors activated, donned respirators and evacuated the area.

WHAT WENT WRONG:

The Maintenance Procedure requirement was not followed as Contractor's training was less than adequate for personnel regarding the instrumentation piping schemes, and instrument connection diagrams.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Maintenance Contractors are required to implement training or an annual knowledge assessment, with periodic participation of Company Maintenance representatives, specifically focused on instrumentation piping and hookup diagrams on Company equipment in accordance with Company Maintenance procedures.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

HIGH POTENTIAL EVENTS RELATED TO PROCESS SAFETY

In this Section, high potential events reported in the safety performance dataset that were PSE related but not classified as PSE are presented.

AFRICA ONSHORE

DATE: 19 Jan 2025

COUNTRY: Gabon

FUNCTION: Construction

CAUSE: Explosion, fire or burns

ACTIVITY: Maintenance, inspection, testing

PRIMARY LIFE-SAVING RULE: Work authorization

SECONARY LIFE-SAVING RULE: Bypassing safety controls

NARRATIVE:

On January 19, 2025, during restart attempts of the HP-A Gas Compressor following corrective maintenance, an unusually loud noise was heard from the exhaust. Seconds later, the fire alarm was triggered, and it was observed that the exhaust structure had fallen off. Technicians in the vicinity immediately pressed the compressor's emergency stop button and evacuated the facility. Simultaneously, the emergency response team was activated, and the on-scene commander initiated a full facility shutdown using a manual call point near the compressor building. Once the source of the loud noise was confirmed to be the HP-A compressor, with the exhaust lying on the ground, the area was cordoned off, and an investigation was launched. The compressor was manually shut down, and the ESD-1 emergency stop was activated.

WHAT WENT WRONG:

During the starting of gas lift compressor (HP-A), the exhaust pipe of the engine driving the reciprocating compressor broke and fell off after several attempts to start it, subsequently activating the fire alarm.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

The crew was not very familiar with the model of engine and didn't know how to react after 1 or 2 attempts to restart.

The operating manual was in English not fully understood by the maintenance team.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

AFRICA OFFSHORE

DATE: 07 Jul 2025

COUNTRY: Ghana

FUNCTION: Production

CAUSE: Pressure release

ACTIVITY: Production operations

PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

On 7th July 2025 at 15:03 hrs, a hydrocarbon gas release occurred onboard the FPSO from anti-surge valve (part of the gas compression system). Gas detectors were activated and an Emergency Shutdown Level 2 (ESD-2) was triggered, resulting in a full production shutdown. The Emergency Response Team confirmed stem leakage on the anti-surge valve.

The valve was isolated and inspected. Only remnants of stem packing were found. A temporary in-situ repair was attempted using graphite split and braided rings, but the installation sequence was incorrect. The valve was therefore not leak-tested and sent onshore for further assessment.

WHAT WENT WRONG:

Critical Factor 1: Stem packing was not checked after approximately 500 strokes for load and re-tightened. If this had been done the valve stem leak would not have occurred in the opinion of the investigation team.

Critical Factor 2: None of the documentation provided with the valve identified the requirement to recheck the tightness of the packing as prescribed at approximately 500 strokes of the valve. The valve assembly drawing lacked detailed information, particularly regarding the installation sequence of the packing types and the lantern ring (a spacer within the packing arrangement).

Critical Factor 3: In conjunction with the lack of details within the drawing available, the IM (Inventory Management) numbers used in the Company system did not fully address the requirements of the valve stem packing requirements.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. PV-2030A-02 was sent onshore for an RCA. Evidence found support the root cause of the leak as described above.
 2. The stem packing has been dismantled and rebuilt at contractor facility, who obtained the detailed drawing for packing installation sequence and components. The valve was tested with nitrogen at 373barg whilst submerged in a water bath to prove leak tight. The valve was stroked during this test. The valve is now back in service.
 3. Ongoing inspection of all actuated valves in gas service to identify any weeps/leaks. FLIR camera is using infra red technology to identify any hydrocarbon releases to atmosphere.
- FV in gas export service was remediated by the offshore team
 - There was also another valve within the weep register, PV-2020-05 which started to leak on 13th August. It has been isolated to stop the atmospheric release as this valve is not required during normal operation.
 - No other valves identified with weeps or leaks.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: 18 Mar 2025

COUNTRY: Mauritania

FUNCTION: Production

CAUSE: Pressure release

ACTIVITY: Maintenance, inspection, testing

PRIMARY LIFE-SAVING RULE: Energy isolation

SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

Contractor carrying out repairs on the Scale Inhibitor Injection system, the system was understood to have been depressurised. As the contractor started to release an end cap, the cap was projected downwards under pressure. The cap went directly downwards into a section of plant close to a walkway. This could have had potential to impact personnel.

WHAT WENT WRONG:

Failure to properly depressurise closed circuit.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Learning 1. The process for managing valve movements during a leak test in response to a leak was not robust. The leak testing procedure was not explicit about how to manage such a situation. Also no bridging document between procedure and company Control of Work procedure.

Learning 2. Site team did not verify that the leak test system envelope was fully depressurised. Task fatigue and normalisation of the risk were identified resulting in the Permit Authority deciding to de-prioritise some oversight of the task.

Learning 3. The potential hazard created using 6Mo tubing and 316 stainless steel fittings at tubing geometries outside of manufacturer recommendation was not understood. Knowing this may have influenced decisions around controls such as exclusion zones and proving zero energy at the point of system being worked on.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

ASIA/AUSTRALASIA ONSHORE

DATE: 05 Sep 2025

COUNTRY: Australia

FUNCTION: Production

CAUSE: Pressure release

ACTIVITY: Maintenance, inspection, testing

PRIMARY LIFE-SAVING RULE: Energy isolation

SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

IP1 observed visible bulging of a drain hose and proceeded to disconnect it. Although the associated compressor had been isolated, the hose remained connected to the main drainage manifold, which was linked to 24 compressors.

The hose remained pressurized, and upon disconnection it struck the IP1 on the left hand and lower leg, and subsequently impacting IP2's hand as they attempted to assist. As a result, IP1 and IP2 were injured.

WHAT WENT WRONG:

The task continued despite incomplete isolation: only the compressor was isolated while the hose remained connected to a pressurized shared manifold, the hose's visible bulging signalled residual pressure, and the plan was informally changed without a full risk assessment.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Recognize and respond to deviations, verify isolation across the entire system, and avoid informal adaptations - promote a disciplined "Stop and Verify" culture that empowers pausing and reassessment when conditions change.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 03 May 2025

COUNTRY: Malaysia

FUNCTION: Production

CAUSE: Explosion, fire or burns

ACTIVITY: Maintenance, inspection, testing

PRIMARY LIFE-SAVING RULE: Hot work

SECONARY LIFE-SAVING RULE: Work authorization

NARRATIVE:

During hot work activities to replace process lines, the flange of a service water line was broken. This resulted in a release of water followed by a hydrocarbon/water emulsion. There was no CO/LEL detected and the material was removed by flushing with service water. Despite a very minor, intermittent liquid release (droplets) from the open line, work activities continued to cut an upstream service water line. About 10-15 minutes later, a fire ignited from the open process nozzle.

WHAT WENT WRONG:

Due to inadequate process isolation, hydrocarbon liquid and vapours from one of the process vessels entered the habitat canvas where hot work was executed. This resulted in a fire on the vessel platform.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Ensure positive isolation of process equipment/lines prior to conducting hot work.
- Ensure adequate procedures for hot work (job method statements with sufficient level of detail; adequate decontamination procedures, adequate safe isolation procedures).
- Ensure contractor alignment on high-risk activities as part of PTW process.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

ASIA/AUSTRALASIA OFFSHORE

DATE: 30 Apr 2025

COUNTRY: Australia

FUNCTION: Production

CAUSE: Explosion, fire or burns

ACTIVITY: Production operations

PRIMARY LIFE-SAVING RULE: Other issue – no applicable rule

NARRATIVE:

A 1 inch thermal isolator coupling failed on a line that is used to mount an ultraviolet light flame detector to the hot oil heater. After the coupling failed,

the detector separated, and the released hydrocarbons ignited within the burner chamber. This triggered 2 independent Infrared flame detectors alarming causing a unit shutdown and ultimately cascaded to an emergency shut down (including blowdown). As designed, the local hot oil heater deluge and foam system successfully activated. No personnel were injured from this event.

WHAT WENT WRONG:

1. Routine inspections lacked a defined step to identify internal corrosion or scale accumulation within the air purge line, as well as stress caused by misalignment of thermal isolators or flame eye detectors.
2. The system's design inherently carried a risk due to the use of high temperature polymer thermal isolators, which are susceptible to failure.
3. Original Equipment Manufacturer guidelines were unavailable to reference the required alignment and fit-up specifications for the thermal isolator.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Work instructions incorporating references to engineering specifications and original equipment manufacturer installation requirements, particularly for sensitive or delicate instrumentation.

Integrating high-temperature component degradation into a structured lifecycle management framework that includes defined replacement criteria and condition assessments based on operational exposure, rather than time-based intervals alone.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 06 May 2025

COUNTRY: Brunei

FUNCTION: Production

CAUSE: Struck by (not dropped object)

ACTIVITY: Maintenance, inspection, testing

PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

During station shutdown, while troubleshooting on the Emergency Shut Down (ESD) actuator valve to reconfirm either it's opened or closed for readiness of purging activity by bleeding N2 supply to close the ESD actuator valve. They reintroduced N2 again to the actuator to open the valve to see whether the indicator of ESD valve change to open position. However, the introduction of N2 caused the actuator end cap ejected.

WHAT WENT WRONG:

1. Competency gap in the personnel involved i.e. not trained nor confident in operating actuators with nitrogen gas.
2. Last minute changes and undocumented valve modifications led to risk exposure.
3. Procedure misalignment in operating actuated valves using nitrogen bottle and handling compressed gas cylinders and tanks and not clearly integrated.
4. The regulator used was unfit for the task, reducing pressure reading accuracy and safety.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Operation and turnaround crew should be trained on operating actuated valves using nitrogen bottle procedure.
2. A separate high-risk (RED) Permit to Work should be provided for the manual operation of actuated valves using nitrogen bottles.
3. Operating actuated valves using nitrogen bottle procedure and handling compressed gas cylinders and tanks procedure should be complementary to each other.
4. Ensure usage of the correct N2 regulator.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 21 Aug 2025

COUNTRY: Thailand

FUNCTION: Construction

CAUSE: Explosion, fire or burns

ACTIVITY: Construction, commissioning, decommissioning

PRIMARY LIFE-SAVING RULE: Other issue – no applicable rule

NARRATIVE:

During Booster compressor (BC) commissioning, the operator observed a smoldering fire in the engine exhaust elbow. The operator immediately shut down the BC unit and extinguished the fire using a portable extinguisher.

WHAT WENT WRONG:

Improper installation of cladding and insulation, where silicone was applied directly on the exhaust elbow and ignited, burning the cladding's plastic coating film.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Establish and implement a pre-installation design review and approval process to ensure that all insulation installation activities are carried out in accordance with approved design. Provide Installers and Inspectors with the reviewed and approved design documents as mandatory reference during installation and inspection.
- Ensure that Contractor's inspectors have a clear understanding of the company's insulation installation procedure and critical acceptance criteria as specified in the Insulation and Personnel Protection Procedure, including lessons learned from past incidents.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

EUROPE OFFSHORE

DATE: 29 Aug 2025

COUNTRY: Norway

FUNCTION: Production

CAUSE: Exposure noise, chemical, biological, vibration, extreme temperature

ACTIVITY: Maintenance, inspection, testing

PRIMARY LIFE-SAVING RULE: Bypassing safety controls

SECONDARY LIFE-SAVING RULE: Work authorization

NARRATIVE:

Two personnel experienced symptoms of severe oxygen deficiency during inspection work, but managed to self-evacuate safely. Subsequent measurements confirmed low oxygen levels inside the enclosure, linked to nitrogen ingress from the turbine seal system combined with a lack of ventilation.

WHAT WENT WRONG:

The incident was triggered by nitrogen leakage from the seal system, which under normal ventilated conditions is harmless. However, following a planned emergency shutdown (ESD) and blackstart prior to shutdown period, key barriers were unavailable: ventilation remained off, and the fire dampers which require manual reopening, stayed

closed. Without airflow, oxygen was gradually displaced overnight creating a hazardous atmosphere that posed a serious risk of unconsciousness and fatality.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Assess feasibility of installing fixed oxygen monitoring in the turbine enclosure and similar areas. If not feasible, require mandatory portable gas testing before entry into locations with potential oxygen depletion risk.

Review and update the blackstart procedure to ensure turbine hood ventilation and dampers are restored before the area is reclassified as safe for entry.

Revise PTW requirements for entries into the turbine enclosure during or after special operational states (e.g., blackstart, shutdown, or loss of HVAC). The PTW must account for potential atmospheric changes, with mandatory portable gas measurements under such conditions.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

MIDDLE EAST ONSHORE

DATE: 09 Nov 2025

COUNTRY: Iraq

FUNCTION: Production

CAUSE: Dropped objects

ACTIVITY: Production operations

PRIMARY LIFE-SAVING RULE: Working at height

SECONDARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

While dismantling a scaffold, a contractor accidentally dropped a scaffold board. The board struck a 0.5-inch, 32-bar pressurized tube, causing physical damage to the tube and resulting in a gas leak.

WHAT WENT WRONG:

Instrumentation tube damage during scaffold dismantling.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Implement barriers (nets, toe boards, edge protection) to prevent objects from falling

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 08 Apr 2025

COUNTRY: Qatar

FUNCTION: Production

CAUSE: Pressure release

ACTIVITY: Maintenance, inspection, testing

PRIMARY LIFE-SAVING RULE: Energy isolation

SECONDARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

A contractor mechanical field service representative and mechanical team had disconnected the export gas compressor gas turbine's primary and secondary flexible fuel gas hoses. After disconnecting the hoses, the contractor field service representative was on the back side of the gas turbine while the mechanical technicians were on the front side. While lying beneath the gas turbine to check the water wash drain tube under the gas turbine, pressure was released, causing the loose hoses to swing.

WHAT WENT WRONG:

One of the loose hoses struck the helmet of the mechanical field service representative, resulting in physical damage to the helmet.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Trapped pressure must always be safely discharged before disconnecting components, even when the system appears isolated - especially in high-pressure gas systems.
- Contractor activities must be fully integrated into company isolation procedures, including proper documentation in Process Isolation Certificates, to avoid oversight of critical steps.
- Safe Work Method Statements (SWMS) must include detailed, step-by-step instructions, particularly for non-routine tasks, to ensure consistent understanding and execution.
- Non-routine activities require Task Risk Assessments (TRA) regardless of pressure levels or perceived risk, as they often involve unexpected hazards.
- System manipulations (e.g., Programmable Logic Controller forcing) must not occur without coordination and approval from the Area Authority, and only after confirming all personnel are in safe positions.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: 21 Jul 2025

COUNTRY: Qatar

FUNCTION: Production

CAUSE: Pressure release

ACTIVITY: Maintenance, inspection, testing

PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

A gas leak was detected from the BV bypass of the liquid filter due to a flange plug snapping out. The leak was successfully stopped, the area was declared safe, and all necessary safety protocols were followed. Operations resumed as normal. No injuries or damage/loss occurred as a result of this release.

WHAT WENT WRONG:

1. The end-blinds utilized were designed with threaded plugs typically intended for use in depressurized systems or during hydrostatic testing. These end-blinds were the only type available at the time of the maintenance activity.
2. No formal engineering guidelines or procedural controls were in place to govern the appropriate application of such end-blinds in pressurized systems.
3. Acceptance of using end-blinds with threaded plugs was identified as the Immediate Cause of the incident.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Establish a comprehensive maintenance procedure that includes an inventory of various classes of blinds, their intended applications, and the approval workflow for blind selection during maintenance planning.
2. Revise the Torquing Procedure to incorporate clear guidance on:
 - a) Conditions under which torquing is mandatory.
 - b) Verification and documentation of applied torque values.
 - c) Competency requirements and qualification criteria for outsourced personnel performing torquing operations.
3. Develop a Hot Bolting Maintenance Procedure detailing scope, risk assessment requirements, applicability, operational controls, and competency standards to ensure safe execution under live conditions.
4. Review of Permit to Work (PTW) Management Practices with the aim to enhance control over carry-over jobs and ensure all changes are properly documented and managed.
5. Reassess the PTW workload assigned to Permit Controllers.
6. Implement best practice guidelines to define the maximum number of permits a controller can manage concurrently or within a single shift.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

MIDDLE EAST OFFSHORE

DATE: 25 Jul 2025

COUNTRY: Qatar

FUNCTION: Production

CAUSE: Pressure release

ACTIVITY: Maintenance, inspection, testing

PRIMARY LIFE-SAVING RULE: Energy isolation

SECONDARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

IP was working alone and attempted to disconnect a 2" airline which was pressurized. An assumption was made that line was de-pressurized without being verified - line was still connected to an air compressor. Upon release, hose kicked back striking IP's hand and helmet.

WHAT WENT WRONG:

The air hose was disconnected while still pressurized because isolation was assumed rather than verified.

Pressurized hose not marked up as per checklist. Line was not walked as per procedure to verify status of line. The hose disconnected was a supply line, not an outlet, due to misidentification of the air distribution configuration. Key hazards (pressurized system, line of fire, verification steps) were not identified before the task, and safe disconnection steps were not followed.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

All equipment shall be clearly labelled and tagged in accordance with site standards across all platforms and operational locations.

Permit-to-work processes shall be reviewed to ensure all hazards and controls are comprehensively identified, including the use of task-specific permits for rigging and de-rigging activities, with controls clearly defined and non-generic.

Undertake formal follow-up on the competency assurance of personnel involved in isolation and permit-related activities, and ensure that any identified competency gaps are addressed in accordance with established training and authorization requirements.

Permits involving pressurized air systems shall require verification of air distribution arrangements, including confirmation that approved air distribution network drawings are attached where applicable.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

NORTH AMERICA ONSHORE

DATE: 01 Dec 2025

COUNTRY: Canada

FUNCTION: Drilling

CAUSE: Pressure release

ACTIVITY: Drilling, workover, well operations

PRIMARY LIFE-SAVING RULE: Other issue – no applicable rule

NARRATIVE:

During preparations to install and test a blowout preventer (BOP) on a re-entry well, a downhole isolation arrangement was in place and surface checks indicated no pressure. While the work sequence was progressing toward BOP pressure/function testing, the well flowed and released steam and reservoir fluids to surface. Personnel evacuated the immediate area and the well was shut in using the BOP blind rams; the release lasted approximately four minutes.

WHAT WENT WRONG:

A downhole isolation barrier (permanent bridge plug) was assessed as having likely lost integrity, allowing reservoir fluids/energy to migrate above the barrier. The sequence of work progressed with limited confirmation of barrier

performance under the evolving conditions, and early indications (e.g., hiss/fluid observations) did not trigger a hold point prior to the flow event.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Strengthen re-entry barrier requirements by adding a verified surface mechanical barrier and an explicit pressure-test requirement prior to continuing critical steps. Incorporate clear “stop/hold” triggers for abnormal indications during well preparation and ensure barrier readiness/verification steps are completed before exposure-increasing activities.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 10 Jul 2025

COUNTRY: USA

FUNCTION: Drilling

CAUSE: Struck by (not dropped object)

ACTIVITY: Drilling, workover, well operations

PRIMARY LIFE-SAVING RULE: Energy isolation

SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

A Mechanic and an E-Tech were repairing a section of the natural gas manifold on frac pump. The employees were replacing a section on the manifold referred to as a pistol. After the repairs were made the employees opened the natural gas supply valve to supply the gas manifold on the pump. The mechanic stepped up onto a platform to open the main ball valve of the natural gas manifold to allow the pump to start burning the natural gas as fuel. As the mechanic (IP) opened the main ball valve, both employees heard a loud pop noise. The top of the low-pressure regulator (VDR) broke away from its base and struck the mechanic in the head above his right eye causing a laceration. The top of the VDR is made of aluminum and weighs approximately 1lb/~454g. The IP was evaluated at a local emergency room, and the IP received 8 sutures to close the laceration.

WHAT WENT WRONG:

No administrative controls to perform LOTO on the pump for this task.

Worker was new to position (little over 3 months) and never worked on gas train.

Safeguard was not in place and caused the system to be over pressured at 115psi. The OEM rating is 5 psi.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

All pumps without complete gas trains were placed out of service and the lock out tag out procedure put into place.

Training was given to shop & test stand employees on the recognition of a complete gas train and procedure for testing regulators before leaving the test stand and putting the pump into service.

E-Tech manager will provide a procedure for the established gas train and the testing procedure in an SOP for running gas including all hookup and parts that are approved.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

DATE: 13 Jun 2025

COUNTRY: USA

FUNCTION: Production

CAUSE: Explosion, fire or burns

ACTIVITY: Production operations

PRIMARY LIFE-SAVING RULE: Other issue – no applicable rule

NARRATIVE:

A manual emergency shutdown was initiated following a report of a valve flange leakage in a relief line to the flare header. Investigation after shutdown concluded that the flange fasteners were under-torqued.

WHAT WENT WRONG:

Flange tightness was not meeting the requirements causing the gas release.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Proper flange management (proper flange assembly, correct torquing) including post-maintenance leak checks are essential to confirm joint integrity.
- Periodic inspection of critical flange joints using hand-held gas detectors and other tools for leak detection.
- Prompt operator response if leaks are detected. Be extra cautious whenever gas is suspected, even without fixed-detector alarms.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

SOUTH & CENTRAL AMERICA ONSHORE

DATE: 16 Feb 2025

COUNTRY: Brazil

FUNCTION: Drilling

CAUSE: Struck by (not dropped object)

ACTIVITY: Transport - Land

PRIMARY LIFE-SAVING RULE: Driving

SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

During the movement of the workover rig truck from one well to another, the rear tyre on one of the truck's axles completely detached, resulting in an impact with the injection line of a well, with no visible damage.

WHAT WENT WRONG:

Detaching of a truck tyre.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Review the maintenance plan to ensure actions that guarantee the compliance of all wheel hub components.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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The Process Safety Event data presented in this Report are based on voluntary submissions from participating IOGP Member Companies and are not necessarily representative of the entire upstream oil and gas industry.

The Process Safety Events (PSE) data presented are based on the numbers of Tier 1 and Tier 2 PSE reported by participating IOGP Member Companies, and are categorized by:

- onshore and offshore
- drilling and production
- activities
- consequences
- material released

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